

Identification of nonsense-mediated decay inhibitors that alter the tumor immune landscape

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Abstract

Summary

Despite exciting developments in cancer immunotherapy, its broad application is limited by the paucity of targetable antigens on the tumor cell surface. As an intrinsic cellular pathway, nonsense-mediated decay (NMD) conceals neoantigens through the destruction of the RNA products from genes harboring truncating mutations. We developed and conducted a high throughput screen, based on the ratiometric analysis of transcripts, to identify critical mediators of NMD. This screen implicated disruption of kinase SMG1's phosphorylation of UPF1 as a potential disruptor of NMD. This led us to design a novel SMG1 inhibitor, KVS0001, that elevates the expression of transcripts and proteins resulting from truncating mutations *in vivo* and *in vitro*. Most importantly, KVS0001 concomitantly increased the presentation of immune-targetable HLA class I-associated peptides from NMD-downregulated proteins on the surface of cancer cells. KVS0001 provides new opportunities for studying NMD and the diseases in which NMD plays a role, including cancer and inherited diseases.

One Sentence Summary

Disruption of the nonsense-mediated decay pathway with a newly developed SMG1 inhibitor with *in-vivo* activity increases the expression of T-cell targetable cancer neoantigens resulting from truncating mutations.

eLife assessment

Here, the authors developed a cell-based screening assay for the identification of small molecule inhibitors of nonsense-mediated decay (NMD), and used it to validate KVS0001, a new small molecule SMG1 kinase inhibitor derived from the existing inhibitor SMG1i-11, showing it inhibits NMD in cultured cells leading to expression of neoantigens from NMD-targeted genes and slows tumor growth of cancer cell lines possessing a significant number of out-of-frame indel mutations. The conclusions are supported by **convincing** evidence, and the significance of this work consists in the development of a new and very promising NMD inhibitor drug that acts as an inhibitor of the SMG1 NMD kinase and is effective in animal tumor studies. This is an **important** advance for the field, as previous NMD inhibitors were not specific, lacked efficacy, or were very toxic and hence not suitable for animal applications.

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Introduction

Despite success with cancer immunotherapies, approved immunotherapies are not available for the majority of cancer patients and only a minority of treated patients realize a durable response[1-4]. Current studies largely focus on discovering new agents and identifying patients most likely to benefit from existing immunotherapies[5]. While many studies correlate tumor insertion and deletion (indel) mutation load with immunotherapeutic response, not all tumors with high indel mutational loads respond to checkpoint inhibitors[6-13].

The typical adult solid tumor contains a median of 54 coding somatic nucleotide variants, many of which have the potential to create novel neoantigens or Mutation Associated NeoAntigens (MANA) [14-16]. Approximately 5% of mutations are insertions/deletions (indels) or splice site changes that alter the open reading frame (ORF) of the transcript[14-16]. This subset of mutations is of particular interest because they can result in proteins and derived peptides that are foreign to a host's healthy cells, giving rise to neoantigens[17-19]. In normal cells, nonsense-mediated decay (NMD) plays an important role in messenger RNA (mRNA) quality control, as well as normal gene expression[20-23]. In cancer cells, however, NMD may aid immuno-evasion by eliminating RNA transcripts coming from genes that carry truncating mutants[19]. This prevents translation and presentation of peptides from these proteins on MHC class I complexes, rendering them invisible to immune cells[24]. Previous work has shown knockdown of the NMD pathway with siRNA enhances the anti-tumor immune response[18, 25].

Analysis of cell lines with loss of the UPF1 RNA helicase and ATPase gene (*UPF1*), a key mediator of NMD degradation, also showed increased levels of aberrant transcripts and mutant proteins in the alleles targeted by NMD[26]. This is reminiscent of the pharmacological modulation of splicing, which has also been shown to increase the number of neoantigens present on the cancer cell surface due to similar underlying mechanisms[27]. Despite early studies showing little toxicity with NMD inhibition, there is as of yet no reports of a specific chemical inhibitor of the pathway with good bioavailability[28-32].

In this work we create a cell-based high throughput assay to query the effects of a curated library of small molecules on NMD function. From this screen, we identify a lead compound capable of inhibiting NMD and ascertain its protein target as nonsense mediated mRNA decay associated

PI3K related kinase (SMG1), a kinase that is a critical mediator of NMD. As the lead compound produced unacceptable toxicity in animal models, we then designed a specific and bioavailable small molecule inhibitor of SMG1 (KVS0001) that is well tolerated *in vivo*. We then demonstrate that targeted inhibition of SMG1 by KVS0001 leads to the presentation of novel neoantigens identifiable by T cells leading to tumor growth inhibition *in vitro* and *in vivo*.

Results

Development of a high throughput assay to identify NMD inhibitors

To develop an assay to find NMD inhibitors, we identified isogenic cell lines with out-of-frame indel mutations, hereinafter referred to as truncating mutations, targeted by NMD activity (Table S1). We previously reported a panel of non-cancerous cell lines in which 19 common tumor suppressor genes were inactivated using the Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR)-Cas9 system[33]. Through evaluation of this panel, we discovered two genes, Stromal Antigen 2 (*STAG2*) and Tumor Protein p53 (*TP53*), which did not express their expected proteins when assessed by western blots (*STAG2*) or immunohistochemistry (p53) (Figure S1). Because the inactivation of these two genes was the result of frame-shift mutations, we suspected that the absence of the proteins was due to NMD. This suspicion was supported by whole transcriptome RNA-sequencing (Figure S2)[33]. Notably we saw an average decrease of *STAG2* RNA transcripts by 20-fold and *TP53* by 6-fold relative to their respective wild-type transcripts in the parental cell lines.

We selected two *STAG2* knockout clones (clone 2 and 8) and one *TP53* knockout clone (clone 221) derived from the Retinal Pigmented Epithelial (RPE1) cell line to design a high-throughput screen (HTS). Treatment with the canonical protein synthesis inhibitor, working indirectly to inhibit NMD, emetine demonstrated up to a 60-fold recovery of mutant RNA transcript expression, establishing this panel as appropriate for HTS (Figure S3)[34].

We then designed a Next Generation Sequencing (NGS) assay for NMD (Figure 1A). We mixed the three cell lines in equal proportions and plated the mixture in 96-wells plates followed by treatment with one compound (drug) per well. We determined NMD inhibition efficacy by comparing the wild-type and truncating mutant transcript expression levels in a ratiometric manner. Specifically, the wild-type sequences of the reciprocally knocked out clone (i.e., wild-type *STAG2* sequence from the *TP53* knockout clones, and wild-type *TP53* sequence from the *STAG2* knockout clones) served as internal references, providing a ratiometric assay of mutant to wild-type transcript abundance. This ratiometric assay minimized confounders introduced by nonspecific transcriptional activators or generally toxic agents.

The use of three cell lines with different truncating mutations from two different target genes minimized the possibility that drugs identified in the HTS were cell-line clone or mutation-specific. Note that the use of these cell lines, carefully mixed, banked, and preserved, did not substantially increase the amount of time or work required to screen a single cell line. A combination of well and plate barcodes allowed the pooling and scoring of over 1920 assays in a single NGS lane (see methods).

Execution of a high throughput screen to identify NMD inhibitors

Previous human clinical trials suggest that off-target toxicity at doses required for NMD inhibition makes emetine and other well-known NMD inhibitors unsuitable for human use[35-39]. We performed a high throughput screen (HTS) to identify more specific NMD inhibitors by treating the isogenic cell line panel described above with a commercially available library consisting of 2,658

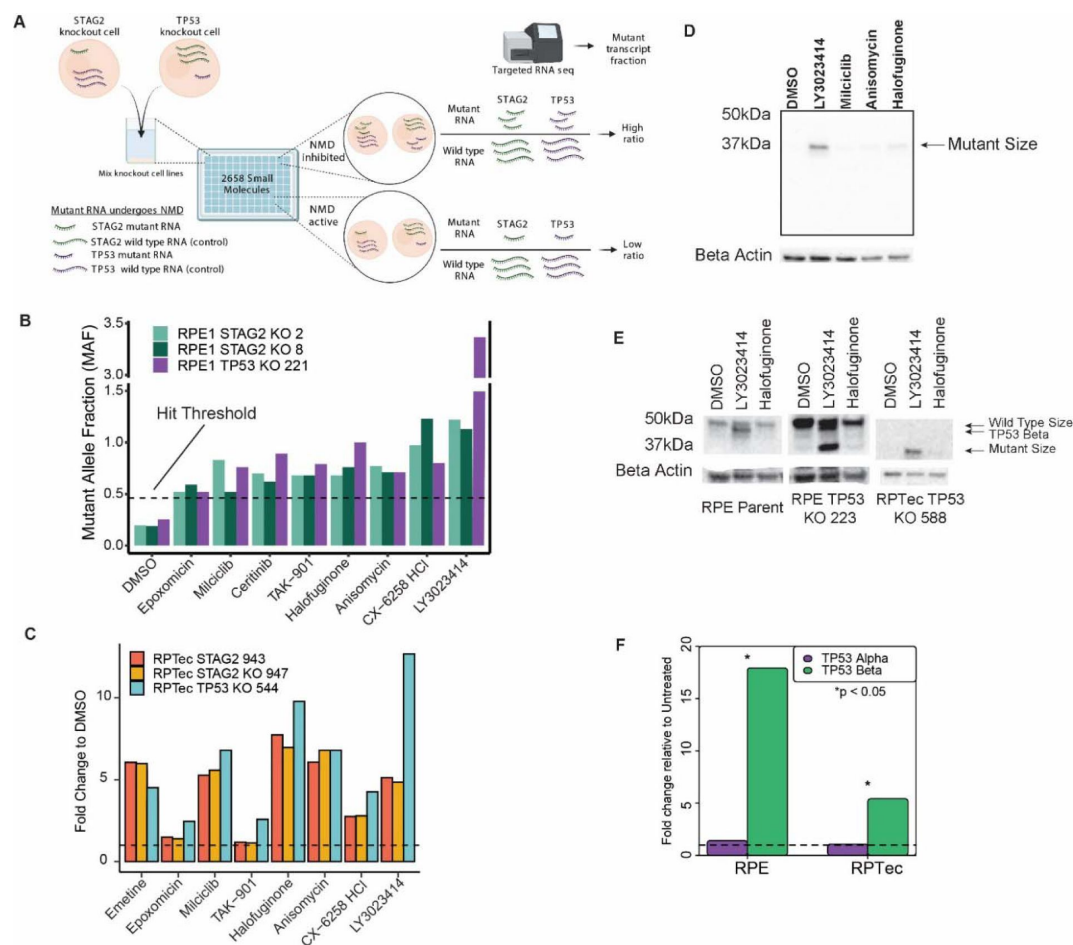


Figure 1

LY3023414 is a small molecule capable of increasing transcription of NMD targets.

A) Schematic of HTS used to identify inhibitors of NMD. Mutant transcripts are represented by a smaller length in the cartoon for illustrative purposes only. All small molecules were tested at 10μM. **B)** Mutant RNA reads relative to wild-type reads for the top 8 hits from the HTS. The dotted line represents the minimum fraction required to be considered a hit (> 5 standard deviations above DMSO control). Full screen results are presented in [Figure S4](#). **C)** Targeted RNA sequencing results of isogenic RPTec knockout clones treated with the 8 best hits from the HTS at 10μM. The dotted line represents a relative RNA expression level of 1, equal to that of DMSO treated wells. Data for ceritinib, which did not validate on any line, is presented only in [Figure S8](#). **D)** TP53 western blot on RPE TP53 224, containing a homozygous *TP53* mutation, using the four hit compounds that validated in RPTec isogenic lines at 10μM. **E)** Western blot analysis of full-length TP53α and isoform TP53β after treatment with two NMD inhibitor lead candidates at 10μM. TP53β (expression known to be controlled by NMD) as well as mutant TP53 are prominently induced by LY3023414 whereas full-length is not. Note that RPE TP53 223 is a heterozygous knockout clone with one near wild-type allele whereas RPTec TP53 588 contains a homozygous *TP53* indel mutation. **F)** qPCR showing 10μM LY3023414 treatment causes increased expression of the NMD controlled alternative transcript for TP53, TP53β, in parent cell lines for RPE1 and RPTec. Significance determined by Student's t-test. Unless indicated otherwise cells were exposed to test compound for 16 hours.

FDA-approved or in late-phase clinical trial small molecules and natural products (Table S2). After purifying RNA from cells 16 hours post-treatment, we scored NMD inhibition using the strategy described in [Figure 1A](#) and [Figure S4](#). Predictably, emetine increased the relative expression of mutant to wild-type transcripts by 3 to 4 fold on average ([Figure S5](#)). Eight compounds (0.3% of the library) increased the ratiometric mutant transcript fraction more than five standard deviations above the DMSO controls in all three cell lines ([Figure 1B](#)). This hit threshold was chosen as it was the minimum required to ensure no false positives were observed in the DMSO controls. One of these eight compounds, anisomycin, is a known inhibitor of protein synthesis and commonly used NMD inhibitor for *in vitro* studies, providing independent validation of the screen[34]. The other seven compounds increased mutant RNA transcript levels 5 to 10-fold relative to untreated cells.

LY3023414 inhibits NMD and causes re-expression of mutant RNA and protein

To validate the eight hit compounds described above, we tested their effects in additional lines with mutations targeted by NMD (Table S1). First, we assessed them on isogenic *STAG2* and *TP53* knockouts in RPTec cells, another non-cancerous cell line (Figures S6A, S6B, and S7)[33]. Four of the original eight hit compounds increased mutant RNA expression in a dose-dependent manner ([Figures 1C](#) and S8). Next, we examined the effects of these four compounds on additional RPE1 *TP53* knockout cell lines with different mutations predicted to generate truncated *TP53* proteins (Figures S6C and S7). While treatment with all four potential NMD inhibitors restored expression of the truncated mutant *TP53* proteins, two of them (LY3023414 and halofuginone) did so most robustly ([Figures 1D](#) and [1E](#)).

Additionally, *TP53* has an isoform, *TP53β*, whose expression is known to be controlled by the NMD pathway[40]. Using cell lines with intact *TP53β* isoform transcripts, we observed an increase in the *TP53β* isoform in cell lines treated with both LY3023414 and halofuginone ([Figure 1E](#) middle and left column). Quantitative real time PCR (qPCR) of both the full-length *TP53* (*TP53α*) and NMD sensitive (*TP53β*) transcripts in parental RPE1 and RPTec cells treated with LY3023414 or halofuginone confirmed the selective upregulation of *TP53β*, but not *TP53α*, transcripts ([Figure 1F](#)). Based on a consistently stronger effect of LY3023414 over halofuginone across multiple isogenic cell lines and assays, LY3023414 was chosen to be the initial lead compound for further studies involving NMD inhibition.

LY3023414 increases expression of NMD repressed mutant RNA transcripts and proteins *in vitro* and *in vivo*

To evaluate whether LY3023414 could relieve NMD repression of naturally occurring heterozygous mutant transcripts, we chose the NCI-H358 and LS180 cancer cell lines (Table S1). Treatment with LY3023414 followed by whole transcriptome RNA sequencing revealed increased expression of the mutant allele in 42% and 67% of heterozygous, out-of-frame, indel mutations in these two lines ([Figure 2A](#)). A third of the nonsense mutations in NCI-H358 were also “recovered” (i.e., mutant transcripts increased relative to wildtype transcripts) after treatment with LY3023414 ([Figure 2A](#) middle). In LS180 there was only one nonsense mutation meeting the required coverage, so it could not be evaluated in depth. Single base pair substitutions not resulting in stop codons were not affected by LY3023414 in either line ([Figure 2A](#) right).

The whole transcriptome sequencing data was confirmed by targeted deep RNA sequencing of two naturally occurring heterozygous truncating mutations from each cell line. Increased expression of the truncated mutant alleles relative to the wild-type alleles of the *EXOC1* and *SPTAN1* genes was observed in NCI-H358 cells after treatment with LY3023414 ([Figure 2B](#) left). Likewise, increased expression of *HNRNPA3* and *LRPPRC* mutant transcripts was observed in LS180 after LY3023414 treatment ([Figure 2B](#) right). We noted no significant effects on two coding region

cachexia and near total inactivity, precluded the longer-term dosing required for anti-tumorigenic effects of LY3023414 in both BALB/c and C57BL/6N immunocompetent mouse strains.

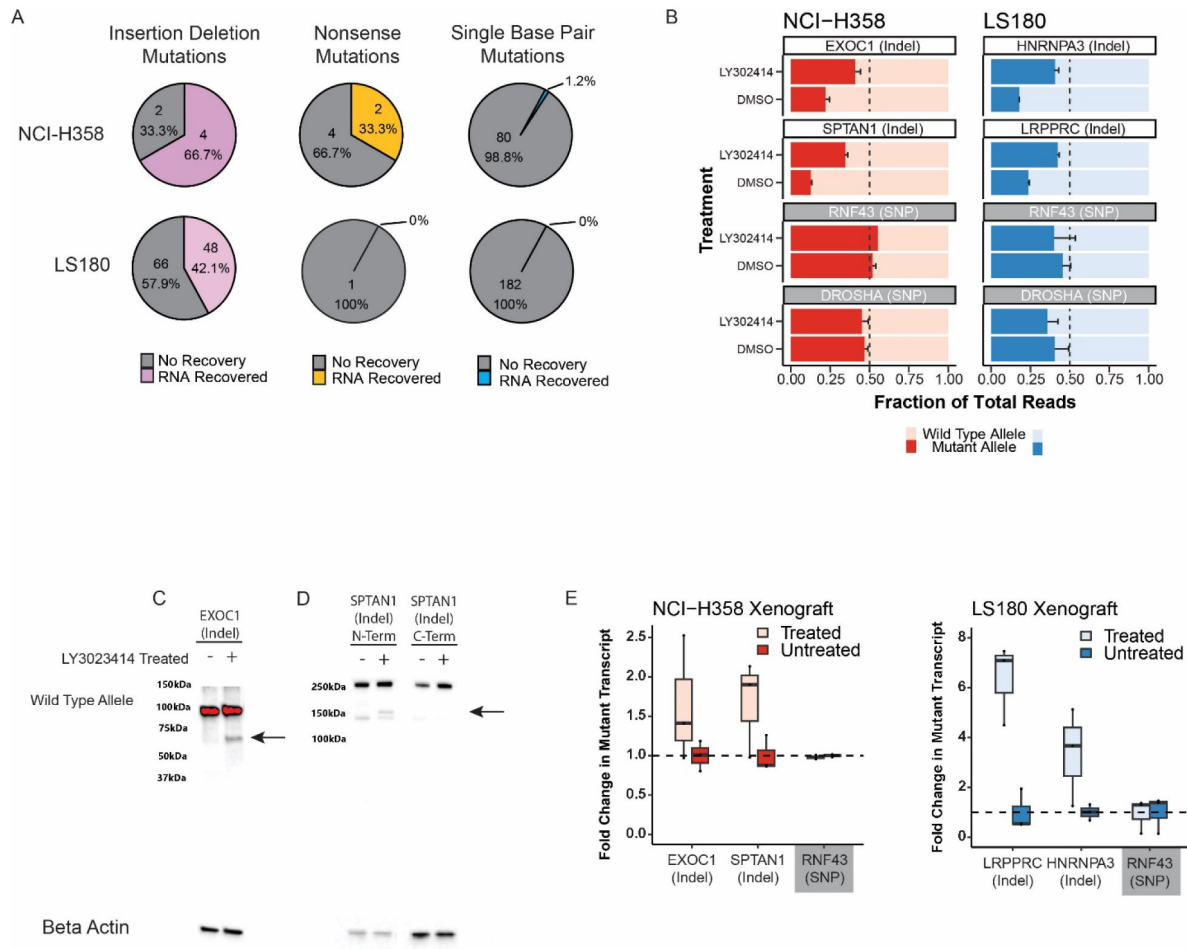


Figure 2

Inhibiting NMD in cancer cells increases broad expression of truncated gene mRNA and protein.

A) Mutant transcript recovery rates for genes containing heterozygous indel mutations based on RNA-sequencing results in cell lines treated with 5 μ M LY3023414 for 16 hours. Strict inclusion criteria were used, such that only mutations with sufficient sequencing coverage are shown (see methods). Recovery is defined as at least 2-fold increase over DMSO treatment. **B)** Targeted high coverage RNA-sequencing confirms recovery of mutant transcript levels in NCI-H358 and LS180 cancer cell lines treated with 5 μ M LY3023414. *RNF43* and *DROSHA* contain common heterozygous SNPs and the mutant allele refers to the non-reference genome allele. Error bars indicate 95% confidence limits. **C)** Western blot analyses of NCI-H358 cells showing mutant and wild-type protein levels in EXOC1 and **D)** SPTAN1 with and without 5 μ M LY3023414 treatment. The black arrow indicates the expected size of the mutant protein. The C-terminal SPTAN1 antibody is downstream of the out-of-frame indel mutation and is not expected to identify the mutant allele. **E)** Fold change in the number of mutant RNA transcripts from deep-targeted RNA sequencing of heterozygous mutated genes in NCI-H358 and LS180 xenografts treated by oral gavage with 60mg/kg LY3023414 assayed 16 hours post-treatment. Student's t-test for target genes are all $p < 0.05$, while the null hypothesis holds for *RNF43* (common SNP).

heterozygous single nucleotide polymorphisms (SNP's) after LY3023414 treatment in the genes *RNF43* or *DROSHA* in either cell line (**Figure 2B** [↗](#)). To substantiate these transcriptomic effects at the protein level, we treated NCI-H358 cells for 24 hours with LY3023414. Western blotting revealed truncated mutant protein for both EXOC1 and SPTAN1 after treatment with LY3023414, whereas the full-length protein levels remained present and unchanged in both the treated and control (DMSO) samples (**Figure 2C** [↗](#) and **2D** [↗](#) left) [\[41\]](#). An antibody that recognizes the C-terminus of SPTAN1, encoded downstream of the frameshift mutation, did not detect mutant protein, as expected (**Figure 2D** [↗](#) right).

LY3023414 was originally developed as a phosphatidylinositol 3-kinase (*PI3K*) inhibitor with activity against AKT Serine/Threonine Kinase 1 (*AKT1*) and Mammalian Target of Rapamycin (*mTOR*) [\[42\]](#). It was tested in a number of clinical trials and has a well-known pharmacokinetic profile *in vivo* [\[43–47\]](#). To determine whether LY3023414 affects NMD *in vivo*, we established xenograft tumors of both NCI-H358 and LS180 in nude mice. Treatment of these mice with a single oral dose of 60 mg/kg LY3023414 led to a significant increase in the expression of mutant RNA transcripts relative to wild-type transcripts 16 hours later (**Figure 2E** [↗](#)). The *RNF43* gene, which harbors a coding region heterozygous SNP, served as a control (**Figure 2E** [↗](#)). Severe drug-associated toxicity, including weight loss bordering on cachexia and near total inactivity, precluded the longer-term dosing required for anti-tumorigenic effects of LY3023414 in both BALB/c and C57BL/6N immunocompetent mouse strains.

The kinase SMG1 is the target for NMD inhibition by LY3023414

To investigate the mechanism of NMD inhibition by LY3023414, we evaluated the six kinases with the highest reported inhibition by LY3023414 [\[42\]](#). siRNA-mediated knockdown of each of these kinases in NCI-H358 and LS180 cancer cell lines was performed for this purpose (**Figure S9** [↗](#)). Only knockdown of *SMG1* resulted in significant changes in the amount of truncating mutant transcript relative to wild-type transcript in all four genes evaluated (**Figure 3A** [↗](#) top). *RNF43* and *DROSHA* contain heterozygous coding region SNP's and served as controls in these experiments and as expected showed no changes despite siRNA treatment (**Figure 3A** [↗](#) bottom).

SMG1 is known to regulate the NMD pathway by activating *UPF1*, an enzyme with RNA helicase and ATPase activity [\[48, 49\]](#). We therefore knocked down *UPF1* with siRNA and found that it restored expression of the NMD-downregulated transcripts, at levels similar to those observed after the knock-down of *SMG1* (**Figure 3B** [↗](#)). Additionally, treatment of NCI-H358 and LS180 cancer cell lines with a previously reported *SMG1* specific small molecule inhibitor, SMG1i-11 resulted in specific increases in the transcripts and proteins from genes with truncating mutations, just as did LY3023414 (Figures S10 and S11) [\[32\]](#). Although SMG1i-11 displayed considerable *SMG1* specificity and marked inhibition of NMD, it was highly insoluble. While we had no difficulty getting it into solution in DMSO for *in vitro* work, we were unable to find a vehicle to administer it *in vivo*, despite numerous attempts at various formulations and administration routes. This may explain why SMG1i-11 has been demonstrated to be an effective *SMG1* inhibitor *in vitro*, but no peer-reviewed reports of its *in vivo* activity have been reported to date [\[32, 50, 51\]](#).

Development of an improved NMD inhibitor targeting SMG1

Although we were unable to secure a viable lead compound, the HTS did identify *SMG1* as an ideal target to disrupt the nonsense-mediated decay pathway. We sought to develop a new *SMG1* inhibitor based on the cryo-electron microscopy structure of the binding pocket of *SMG1* [\[52, 53\]](#). We attempted the synthesis of eleven compounds (KVS0001 to KVS0011) and tested for bioavailability and preservation of target specificity. Among these, KVS0001 stood out due to its solubility while preserving *SMG1* inhibitory activity (Table S2, **Figures 3C** [↗](#) and S12). Mass spectrometry-based assays showed that KVS0001 inhibits *SMG1* protein more than any of the other 246 protein or lipid kinases tested at concentrations from 10nM to 1μM (Table S4 and **Figure**

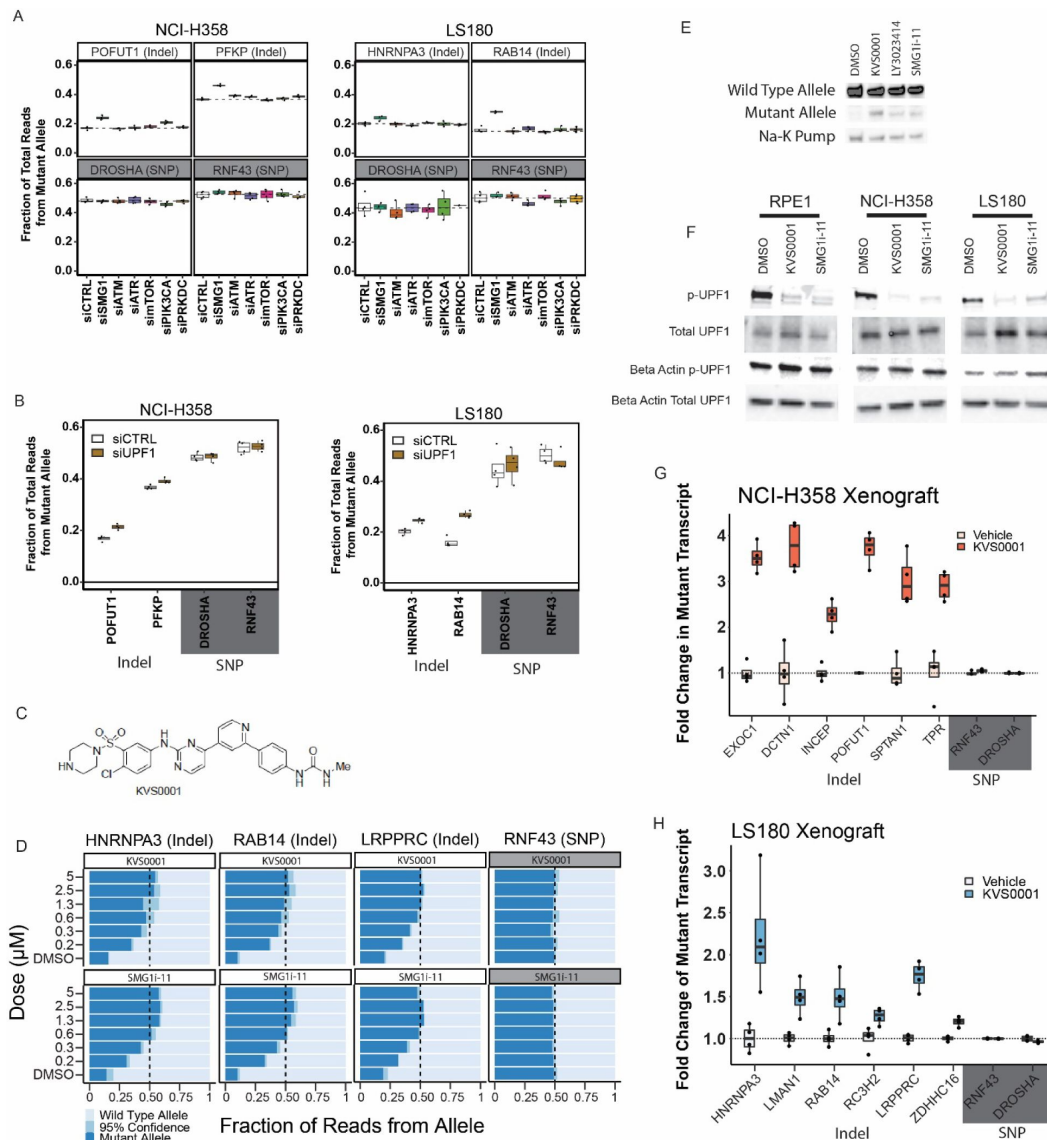


Figure 3

Novel NMD inhibitor KVS0001 is SMG1 specific and induces expression of NMD targeted genes *in vitro* and *in vivo*.

A) Fraction of mutant allele transcripts in genes with heterozygous indels previously established in this study as sensitive to NMD inhibition. Results show mutant levels after siRNA treatment targeting kinases inhibited by LY3023414. *RNF43* and *DROSHA* are common heterozygous SNPs (shaded gray) and serve as negative controls. **B)** Fraction of mutant allele transcripts in genes with truncating mutations known to be sensitive to NMD inhibition after siRNA treatment with siUPF1 or non-targeting siRNA. Data from deep-targeted RNA-sequencing. **C)** Structure of novel NMD inhibitor KVS0001. **D)** Targeted RNA-sequencing on three genes with heterozygous, out-of-frame, indel mutations in LS180 cancer cells treated in a dose-response with KVS0001 or SMG1i-11. *RNF43* serves as a control (common heterozygous SNP) and the mutant allele refers to the non-reference genome allele. **E)** Western blot of EXOC1 protein in NCI-H358 cells treated with 5 μ M novel inhibitor KVS0001, LY3023414 or SMG1i-11 for 24 hours. **F)** Western blot of phosphorylated UPF1 on three cell lines treated with 5 μ M KVS0001, SMG1i-11, or DMSO. Note that total UPF1 and p-UPF1 were run on different gels, loading controls correspond to indicated gel. **G)** Fold change in the number of mutant allele transcripts measured by targeted RNA-Seq in genes containing heterozygous out-of-frame indel mutations in NCI-H358 or **H)** LS180 subcutaneous xenografts in bilateral flanks of nude mice. Mice were treated once with IP injection of vehicle or 30mg/kg KVS0001 and tumors harvested 16hrs post IP treatment. All genes shown contain heterozygous out-of-frame truncating mutations except *RNF43* and *DROSHA* which serve as controls (contain heterozygous SNP's).

S13 [\[54\]](#). Noteworthy off-target kinase inhibition was not observed until doses of 1 μ M and above. Experiments using NCI-H358 and LS180 cells showed that KVS0001 is bioactive in the nanomolar range and subverts the NMD-mediated down-regulation of truncating mutant transcripts and proteins (**Figures 3D**, [3E](#), [S14](#), [S15](#)). Inhibition at concentrations as low as 600nM led to equal expression of wild-type and mutant transcripts, suggesting near total blockade of the NMD pathway (**Figures 3D**, [S14](#)). Western blotting showed that KVS0001 substantially decreases the amount of phosphorylated UPF1, the downstream target of SMG1 kinase activity, in three different cell lines (**Figure 3F**). Finally, KVS0001 treatment of NCI-H358 and LS180-derived xenograft tumors in nude mice resulted in significant increases in transcript levels in each of six tested endogenous genes with truncating mutations, while having had no measurable effects on genes containing heterozygous coding region SNPs (**Figures 3G** and [3H](#)).

NMD inhibition with KVS0001 causes MHC class I display of hidden neoantigens

Cancer cells may evade immune surveillance by downregulating genes with truncating mutations as a result of NMD [[19](#), [55](#)]. Indeed, previous studies with non-specific or toxic NMD inhibitors have shown an increase in selected antigens from tumor-specific mutations when NMD is inhibited [[56](#)]. We therefore investigated whether KVS0001 could similarly alter the cell surface presentation of proteins from genes harboring truncating mutations in NCI-H358 and LS180 cells. Using quantitative HPLC-mass spectrometry (MS) we observed a striking (45 to 90 fold) increase in the *EXOC1* and *RAB14* derived neoantigens, and a significant (2 fold) increase in the *ZDHHC16* derived neoantigen (**Figures 4A**, [4B](#), [S16](#), [S17](#)) [[57](#)]. This is consistent with the re-expression of mutant transcript and protein shown previously in this study (**Figures 3D** and [3E](#)). These three peptides were chosen based on an *in-silico* review of potentially presented peptides and a preliminary experiment that looked at qualitative (present or not present) presentation of the predicted peptides [[58](#), [59](#)].

Targetable peptide presentation occurs with NMD inhibition by KVS0001

To test whether cancer cell neoantigens presented as a result of NMD inhibition could be targeted, we evaluated two cancer cell lines, NCI-H716 and NCI-H2228 (Table S1). Both lines contain homozygous truncating mutations in *TP53* which produce transcripts down-regulated by NMD, with each exhibiting RNA levels less than 20% of the median level of expression among 675 cancer cell lines (**Figures 4C** and [S18](#)) [[60](#), [61](#)]. Treatment with KVS0001 increased the expression of TP53 in both lines, while the commonly used therapeutic agents 5-fluorouracil and etoposide did not affect TP53 protein abundance (**Figures 4D** and [S19](#)).

To determine whether this disruption of NMD repression is targetable by T-cells, we developed a bispecific antibody (KVS-BI043) that recognizes a peptide-HLA complex on one end and CD3 on the other end. CD3 is expressed only on T-cells, and this bispecific antibody functions as a T-cell engager, linking target cells to cytotoxic T-cells, which then kill the targets [[62](#), [63](#)]. KVS-BI043 recognizes a ten amino acid peptide (residues 125 to 134 of TP53) bound to HLA-A24. NCI-H2228 naturally express A24 whereas NCI-H716 cells were engineered to express A24 using a retrovirus (NCI-H716_A24). Treatment of NCI-H716_A24 or NCI-H2228 cells with KVS0001 and the KVS-BI043 bispecific antibody in the presence of normal T-cells caused a significant increase in interferon (IFN- γ) release (**Figure 4E**). We observed no changes in IFN- γ levels in the absence of KVS-BI043 or normal T-cells (**Figure 4E** grey boxed lanes). Most importantly, treatment of NCI-H716_A24 with KVS0001 in the presence of KVS-BI043 and T-cells led to significant killing of the target cancer cells, which was also not observed in the absence of T-cells or the absence of HLA-24 in the target cells (**Figure 4F**). NCI-H2228 was assessed for killing but expressed too much TP53 at baseline (**Figure 4D** left most lane) and thus displayed substantial killing even in the DMSO controls.

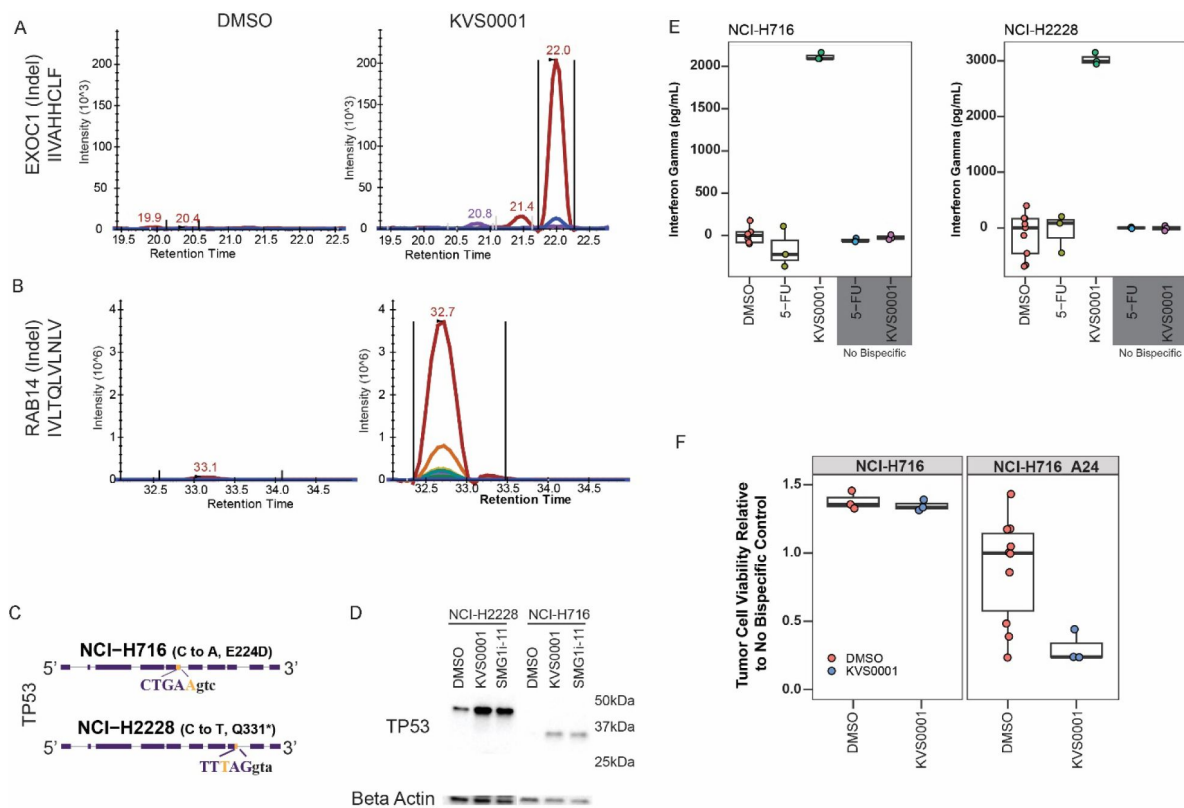


Figure 4

KVS0001 treatment induces targetable cell surface presentation of peptides known to be downregulated by NMD.

A) MHC class I HLA presentation of mutant specific peptide sequences from NCI-H358 and **B)** LS180 cells by quantitative HPLC-Mass Spectrometry treated with DMSO or 5 μ M KVS0001. The gene name, type of mutation (in parenthesis) and presented peptide are shown on the y-axis for each gene. Colors indicate different ions. **C)** *TP53* gene structure and mutant DNA sequence for NCI-H716 and NCI-H2228 cancer cell lines, both contain a homozygous splice site mutation in *TP53*. Capital letters represent exonic sequence; lowercase letters represent intronic sequence. DNA mutation reflected by gold bases. **D)** Western blot against TP53 in the presence or absence of 5 μ M NMD inhibitor in NCI-H716_A24 and NCI-H2228 cell lines. NCI-H2228 has an expected size of 46.6 kDa and NCI-H716 of 34.7 kDa. **E)** IFN- γ levels over baseline based on ELISA in a co-culture assay with NCI-H716_A24 and NCI-H2228 cells, 1.25 μ M NMD inhibitor, human CD3+ T-cells, and bispecific antibody for TP53 and CD3. Chemotherapy (5-Fluorouracil) is shown as a control. **F)** Cell killing based on luciferase levels in a co-culture assay in NCI-H716 cells with and without A24 expression, treated with TP53-CD3 bispecific antibody, 1.25 μ M NMD inhibitor and human CD3+ T-cells.

Tumor growth is slowed in mice treated with KVS0001

Finally, we investigated whether KVS0001 could impact tumor growth in syngeneic models in which the native immune system might play a role (Table S1). For this purpose, we first used murine RENCA (renal cancer) and LLC (lung cancer) as they are known to have a relatively large number of out-of-frame indel mutations (Table S5). Although murine and human SMG1 are highly related (98% at the amino acid level), it was important to show that KVS0001 could actually inhibit NMD in murine cells.

For this purpose, we tested eight genes in LLC, and four in RENCA, which contained out-of-frame indel mutations potentially targeted by NMD. We also assessed the expression of three genes without any mutations known to have their normal expression controlled by NMD[64]. We found that six of the twelve truncating mutation-containing genes and five of the six expression controlled by NMD genes had significantly increased RNA following *in vitro* treatment with KVS0001 (Figure 5A and 5B).

We then implanted LLC and RENCA cancer cells in the mammary fat pad of C57BL/6N and BALB/c mice respectively and treated with KVS0001 or vehicle control (Figure 5C). The dose of KVS0001 was based on experiments showing that the maximum solubility limit was reached around 2 to 3mg/mL, leading to a maximum single dose of 30mg/kg per mouse per treatment. At this dose, the only toxicity noted was transient weight loss (Figure S20), and no other pathology was observed. Both tumor types experienced significant slowing of tumor growth (Figure 5D). However, when the same tumors were implanted in immunocompromised mice, there was no statistically significant difference in tumor growth between mice treated with KVS0001 or vehicle control (Figure 5E).

We evaluated a total of seven syngeneic mouse cancer cell lines. Four contained a relatively high number of truncating indel mutations, while the remaining three contained a moderate or low number. None of the three models with moderate or low numbers of truncating mutations responded to KVS0001 (EMT6, B16-F10, M3 Melanoma) (Figure S21). Of the four models with a higher number of truncating mutations, two (LLC and RENCA) responded in a statistically significant fashion while the other two failed to reach significance (CT26, MC38) (Figures 5D and S21).

Discussion

We report the design and execution of a high throughput screen to identify small molecule inhibitors of the nonsense-mediated decay pathway. The method employs a ratiometric output, allowing quantitative and controlled determination of RNA expression changes associated with NMD. Unlike previous NMD screens, this assay is directed at NMD as a process rather than as a way to identify compounds for treating specific mutations related to a disease state, such as cystic fibrosis or β -thalassemia[28, 30, 65]. The genetically modified isogenic cell lines employed here are uniquely suited for this approach, producing a high signal-to-noise ratio. Follow-up experiments confirmed a low false positive rate in the screen: four of the eight hits re-expressed transcripts from genes with truncating mutations in a dose-responsive manner when re-tested. Note that our assay allows for the detection of NMD inhibitors regardless of effects on protein translation by virtue of the comparison of wild-type to mutant transcripts for each queried gene. This is an important feature, as many previously reported inhibitors of NMD, such as emetine and anisomycin, rely on inhibition of protein translation and thus are unlikely to be optimal for specific restoration of NMD-targeted RNA and protein expression[35, 38, 39, 66]. During the review process it was also noted that different NMD inhibitors caused re-expression of different knockout clones to different extents (see Figure 1B and S8 as an example). It is not

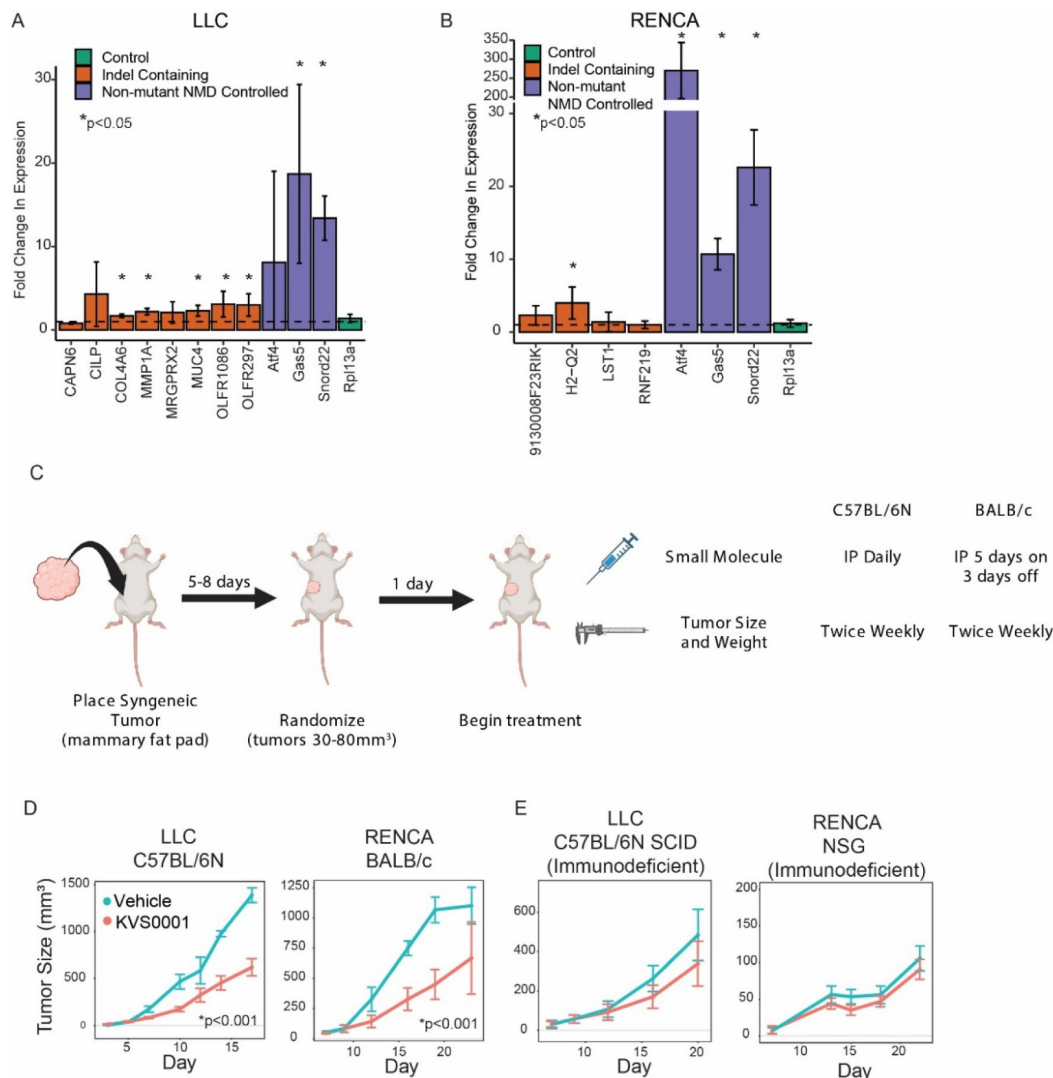


Figure 5

***In vivo* treatment of murine tumors with KVS0001 yield differential tumor growth compared with vehicle treatment.**

A) Fold change in RNA transcript levels in LLC or **B)** RENCA cells treated in-vitro with 5 μ M of NMD inhibitor KVS0001 or DMSO. Orange bars indicate genes with homozygous indel mutations potentially targeted by NMD. Purple bars show genes with no mutations but that are known to have their normal transcription levels controlled by NMD. Green bar is a control gene that should not change with treatment. The dotted line shows relative expression of DMSO treatment (equal to 1). **C)** Treatment schedule for syngeneic tumor mouse experiments. **D)** Average tumor size of LLC (left) and RENCA (right) syngeneic tumors in immune-competent mice (n=8) treated with 30 mg/kg KVS0001 or vehicle control. Difference is statistically significant after day 10 based on one-way ANOVA with Dunnett's test p<0.001 (p<0.05 for day 23 RENCA data point) for both tumors tested. **E)** Average tumor size of LLC (left) and RENCA (right) in immunodeficient mice (n=8) treated with 30 mg/kg KVS0001 or vehicle control. Error bars show 95% confidence intervals in all plots.

clear to us why this occurred, but we speculate that it may be related to differences in the mechanism of action (i.e., protein synthesis inhibition based NMD effects versus targeted NMD pathway inhibition as one example).

The optimal inhibitor of NMD found in our screen was LY3023414, and we suspected that this compound may inhibit the kinase *SMG1* based on previously published kinase activity data with this small molecule[42]. *SMG1* acts as the gatekeeper of *UPF1* activity through its phosphorylation at multiple sites[48]. Unfortunately, issues likely related to off-target (pan-kinase) toxicity at the doses required for NMD inhibition prevented LY3023414 from being utilized in further *in vivo* work. While there are previously described *SMG1* inhibitors, all have undesirable off-target or biophysical characteristics complicating their use as therapeutic agents[28-31, 39, 56, 67]. For example, SMG1i-11 was previously identified as a potent and specific *SMG1* inhibitor[32]. However, in our hands, it was not soluble at concentrations required for *in vivo* work in a suitable vehicle for mouse administration.

Another unrelated compound has shown successful *in vivo* re-expression of mutant RNA transcripts, but the effect is reported as less than a two-fold change, compared to the three to four-fold demonstrated here by KVS0001, and little work has been done to develop the compound further in subsequent studies[68].

Despite these chemistry related challenges, our screen offered an unbiased assessment of the best targets for NMD pathway disruption, and thus we concluded that *SMG1* was an ideal protein to inhibit for this purpose. In light of this, we attempted to improve the biophysical properties of SMG1i-11 through the development of KVS0001. As reported in this study, KVS0001 was a specific inhibitor of *SMG1* that was soluble and could easily be administered to mice. Treatment of KVS0001 not only increased levels of transcripts and proteins from genes with truncating mutations, but also increased the levels of corresponding peptide bound-HLA complexes on the cell surfaces. We developed a new bispecific antibody, KVS-B043, which could recognize these pHLA complexes when the truncating mutations were in the *TP53* gene and destroy cancer cells harboring such mutations. In syngeneic mouse models, KVS0001 slowed the growth of some tumors containing multiple truncating mutations, but only when the mice were immunocompetent. This dependence on an intact immune system supports the idea that the slowed growth of the treated tumors was due to recognition of neoantigens.

Despite these encouraging data, the potential utility of KVS0001 for therapeutic purposes remains speculative, for several reasons. First, we do not understand why KVS0001 (and other NMD inhibitors) increase the expression of many, but not all, genes with truncating mutations. We also do not systematically address the changes to splicing that are undoubtedly occurring when NMD is disrupted in either normal or tumor tissue. More research on the biochemistry of NMD will undoubtedly be required for such understanding, and perhaps the tools described here may facilitate that effort. Second, treatment with KVS0001 reduced the growth of some but not all syngeneic tumor types tested in mice. There appeared to be a correlation between the ability of KVS0001 to inhibit growth and the number of truncating mutations in the tumor. Likewise, recent evidence has been presented showing that cells deficient in known splicing factors are sensitive to subsequent NMD inhibition[69]. This situation is reminiscent of that encountered with immune checkpoint inhibitors, in which there is a positive but imperfect correlation with loss of microsatellite instability (MSI) related proteins and tumor mutation burden[8, 12]. Undoubtedly, the ability of the immune system to react to neoantigens is related to their quality as well as their quantity, with quality defined as the ability of the neoantigen to be bound to the host's particular MHC constitution and initiate an immune response[70, 71]. Third, treatment with KVS0001 led to decreased tumor growth in immunocompetent mice, but did not lead to tumor regressions of the type mandated by RECIST criteria in human clinical trials[72, 73]. Whether this failure to cure tumors in mice results from the contrived nature of the models used –

injection of large numbers of rapidly growing cancer cells into animals with relatively little time to react prior to their demise – complicates the interpretation of many pre-clinical immunotherapeutic approaches.

Finally, KVS0001, though it was soluble and could be administered to animals for at least a month, was not entirely non-toxic, causing transient weight loss in some of the mice. Previous work has raised concerns that the importance of NMD in development and normal gene expression make it a difficult pathway to safely disrupt[21 [↗](#), 74 [↗](#)-77 [↗](#)]. The limited impact on body weight or observed physical activity of mice undergoing KVS0001 therapy at therapeutically relevant doses suggests that NMD inhibition may have acceptable toxicity and be tolerated in developed animals. Another example of this lies in the natural genetic variants of NMD components, which convey variability in the efficiency of NMD between humans, and lends itself to supporting that knockdown of this pathway may be tolerable[78 [↗](#)-80 [↗](#)]. Studies in mice also support this observation, as post-development knockdown of UPF1 displayed minimal phenotype change[64 [↗](#)]. The broad use of NMD in normal development and growth, coupled with the observations here, suggests future work with inhibitors of this pathway should continue with a close eye for on-target off-tissue toxicity.

Though there is much work to be done, we hope that the tools, approaches, and compounds described here will facilitate that work. Should the administration of KVS0001 or a related compound prove non-toxic and well-tolerated in humans, we end with a speculation about the possibility of KVS0001 being used as a preventative rather than as a therapeutic agent. Its potential use to prevent the onset of symptoms in pediatric syndromes caused by germline truncating mutations is obvious[17 [↗](#), 81 [↗](#)]. Less obvious is the potential for it to prophylactically reduce cancer incidence in patients with hereditary non-polyposis colorectal cancer. These patients inherit heterozygous mutations of a mismatch repair gene, and they do not develop tumors until biallelic mutations of that mismatch repair gene are acquired in a rare stem cell during the second or third decade of life. Similar speculations can be made about the potential of KVS0001 to be used to prevent cancer initiation or progression in patients with other inherited mutations in repair genes, or in individuals exposed to high levels of exogenous mutagens.

Acknowledgements

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Author contributions

A.L.C, K.W.K., and N.W. conceptualized the study. A.L.C, S.S., B.V., N.P., C.B., K.G., S.Z., K.W.K., and N.W. developed the methodology. A.L.C., S.S., L.D., E.W., B.P., B.S.L., S.P., E.H., M.P., and N.W. conducted the experiments. A.L.C., S.S., J.D.C., B.S.L., and N.W. performed the formal data analysis. A.L.C, N.W., and K.W.K. wrote the paper.

Declaration of interests

BV, KWK, & NP are founders of Thrive Earlier Detection, an Exact Sciences Company. KWK, NP are consultants to Thrive Earlier Detection. BV, KWK, NP, SZ hold equity in Exact Sciences. BV, KWK, NP, SS and SZ, are founders of or consultants to and own equity in ManaT Bio., Haystack Oncology, Neophore, CAGE Pharma and Personal Genome Diagnostics. NP is consultant to Vidium. BV is a consultant to and holds equity in Catalio Capital Management. SZ has a research agreement with BioMed Valley Discoveries, Inc. CB is a consultant to Depuy-Synthes, Bionaut Labs, Haystack Oncology, Privo Technologies and Galectin Therapeutics. CB is a co-founder of OrisDx and Belay Diagnostics. The companies named above, as well as other companies, have licensed previously described technologies related to the work from this lab but not this paper from Johns Hopkins University. BV, KWK, NP, JDC, and SS are inventors on some of these technologies. Licenses to these technologies are or will be associated with equity or royalty payments to the inventors as well as to Johns Hopkins University. Provisional patent applications on the work described in this paper have been filed by Johns Hopkins University. The terms of all these arrangements are being managed by Johns Hopkins University in accordance with its conflict of interest policies.

Supplemental Materials

STAR methods

Key resource table

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
anti goat 680RD used for Western blot at 1:10K	Licor	926-68074
anti mouse 680RD used for Western blot at 1:15K	Licor	926-68072
anti mouse HRP used for Western blot at 1:2500	Jackson Immuno Research	115-035-006
anti rabbit 800CW used for Western blot at 1:10K	Licor	926-32213
anti rabbit HRP used for Western blot at 1:2500	Jackson Immuno Research	111-035-006
Beta-actin used for Western blot at 1:5000	Cell Signaling Technology	3700S
EXOC1 used for Western blot at 0.4 µg/mL	Abcam	ab251853
LMAN1 used for Western blot at 1:200	ThermoFisher Scientific	CF502137
OKT-3 used for T-Cell Culturing at 15 ng/mL	Biolegend	317347
p21/WAF1 used for Western blot at 1:1000	Cell Signaling Technology	2947S
p53 used for Western blot at 1:1000	Cell Signaling Technology	18032S
phospho-UPF1 used for Western blot at 1:1000	Cell Signaling Technology	2851S
Sodium Potassium Pump (Na-K) used for Western blot at 1:500	Cell Signaling Technology	3010S
SPTAN1 (polyclonal) used for Western blot at 0.04 µg/mL	Bethyl	A301-249A
SPTAN1 (N-Terminal) used for Western blot at 1:1000	Abcam	ab11755
SPTAN1 (C-Terminal) used for Western blot at 1:1000	Abcam	ab75755
STAG2 used for Western blot at 1:1000	Cell Signaling Technology	5882S
UPF1 used for Western blot at 1:2500	Abcam	ab10510
Flow cytometry antibody: α-HLA-A24 used for Flow cytometry at 10 µg/mL	MBL Life Science	Cat #K0208-A64
TP53-CD3 bispecific antibody used for Co-culture at 12.5pg/mL	This Study	N/A
Bacterial and virus strains		
HLA-A24 Retrovirus	Hsiue et al. Science, 2021.	N/A
Luciferase lentivirus	OriGene	Cat #PS100071
Biological samples		
N/A		
Chemicals, peptides, and recombinant proteins		
1% Penicillin-Streptomycin	Thermo Fisher Scientific	Cat #15140122
4–15% Mini-PROTEAN TGX Precast Protein Gels	Bio-Rad	Cat #456-1086
5-Fluoruracil	Sigma Aldrich	Cat #F6627
AMPure beads	Beckman Coulter	Cat #A63880
Cremaphor	Sigma Aldrich	Cat #C5135

DharmaFECT1 transfection reagent	Horizon	Cat #T-2001-02
Dimethyl Sulfoxide (DMSO)	Sigma Aldrich	Cat #C6295
DMEM medium	Gibco	Cat #11995065
EMEM medium	ATCC	Cat #30-2003
Emetine	Sigma Aldrich	Cat #7083-71-8
EPITHELIAL CELL MEDIUM-Complete Kit	Science Cell Research	Cat #4101
Fetal Bovine Serum (FBS)	HyClone	Cat #16777-006
Glycerol	Sigma Aldrich	Cat #G5516
KVS0001	This study	N/A
LY3023414	Selleckchem	Cat #S8322
Matrigel Phenol Red Free Standard Formulation	Corning	Cat #356237
MEGM Mammary Epithelial Cell Growth Medium BulletKit	Lonza	Cat #CC3150
Methylcellulose	Sigma Aldrich	Cat #M6385
Phosphate Buffered Saline (PBS)	Thermo Fisher	Cat # J60465.K2
Phusion Flash High-Fidelity PCR Master Mix	Thermo Fisher	Cat #F548S
Pierce ECL Western Blotting Substrate	Thermo Fisher	Cat #32106
PRIMETIME Gene Expression master mix	Integrated DNA Technologies	Cat #1055770
Protease inhibitor	Millipore Sigma	Cat #4693159001
QIAshredder	Qiagen	Cat #79656
Recombinant IL-2 protein	Prometheus Therapeutics and Diagnostics	Cat #aldesleukin
Recombinant IL-7 protein	BioLegend	Cat #581908
RediJect D-Luciferin Ultra Bioluminescent Substrate	PerkinElmer	Cat #770505
RIPA Lysis and Extraction buffer	Thermo Fisher	Cat #89901
RNA Later	Invitrogen	Cat #AM7020
RPMI 1640 medium	Gibco	Cat #11875-119
Selleckchem Bioactive Compound library (Table S2)	Selleckchem	Cat # L1700
SMG1specific inhibitor: 11j	Ascendex LLC	N/A
SsoAdvanced Universal SYBR Green Supermix	Bio-Rad	Cat #1725270
Trypsin	Gibco	Cat #25300054
Critical commercial assays		
AllPrep DNA/RNA Mini Kit	Qiagen	Cat #80204
Agilent RNA ScreenTape	Agilent	Cat #5067-5576
Agilent RNA ScreenTape Sample Buffer	Agilent	Cat #5067-5577
Agilent RNA Ladder	Agilent	Cat #5067-5578
BCA Protein Assay Kit	Thermo Fisher	Cat #23227
Bio-Rad SingleShot Cell Lysis kit	Bio-Rad	Cat #1725080
ELISA	R&D Systems	Cat #SIF50C
High-Capacity cDNA Reverse Transcription Kit	Advanced Biosystems	Cat #4368814
Illumina RNA library prep kit	Illumina	Cat #RS-122-2001
Kinase kinativ assay	ActivX Biosciences	N/A
Luciferase Assay System	Promega	Cat #E1501
RNeasy	Qiagen	Cat #74104
Deposited data		
Whole transcriptome RNA-seq: NCI-H358	This study	EGAS00001007597
Whole transcriptome RNA-seq: LS180	This study	EGAS00001007597

Whole transcriptome RNA-seq: Isogenic knockout cell lines	Cook, et. al. iScience, 2022.	EGAD00001008559
Experimental models: cell lines		
B16-F10	ATCC	Cat #CRL-6475
HEK293T	ATCC	Cat #CRL-3216
Lewis lung carcinoma (LLC)	ATCC	Cat #CRL-1642
LS180	ATCC	Cat #CL-187
MCF10a	ATCC	Cat #CRL-10317
NCI-H716	ATCC	Cat #CCL-251
NCI-H2228	ATCC	Cat #CRL-5935
NCI-H358	ATCC	Cat #CRL-5807
Renca	ATCC	Cat #CRL-2947
RPE1	ATCC	Cat #CRL-4000
RPE1 STAG2 2	Cook, et. al. iScience, 2022.	N/A
RPE1 STAG2 5	Cook, et. al. iScience, 2022.	N/A
RPE1 STAG2 6	Cook, et. al. iScience, 2022.	N/A
RPE1 STAG2 8	Cook, et. al. iScience, 2022.	N/A
RPE1 TP53 221	Cook, et. al. iScience, 2022.	N/A
RPE1 TP53 223	Cook, et. al. iScience, 2022.	N/A
RPE1 TP53 224	Cook, et. al. iScience, 2022.	N/A
RPTec	ATCC	Cat #CRL-4031
RPTec STAG2 3	Cook, et. al. iScience, 2022.	N/A
RPTec STAG2 943	Cook, et. al. iScience, 2022.	N/A
RPTec STAG2 943A	Cook, et. al. iScience, 2022.	N/A
RPTec STAG2 944	Cook, et. al. iScience, 2022.	N/A
RPTec STAG2 945	Cook, et. al. iScience, 2022.	N/A
RPTec STAG2 946	Cook, et. al. iScience, 2022.	N/A
RPTec STAG2 947	Cook, et. al. iScience, 2022.	N/A
RPTec STAG2 951	Cook, et. al. iScience, 2022.	N/A
RPTec STAG2 952	Cook, et. al. iScience, 2022.	N/A
RPTec STAG2 953	Cook, et. al. iScience, 2022.	N/A

RPTEC TP53 544	Cook, et. al. iScience, 2022.	N/A
RPTEC TP53 588	Cook, et. al. iScience, 2022.	N/A
Experimental models: Organisms/strains		
<i>Mus musculus</i> BALB/cAnAHsd	Harlan Laboratories	
<i>Mus musculus</i> C57BL/6NCrI	Charles River GmbH	
<i>Mus musculus</i> BALB/cAnNCrI	Charles River GmbH	
<i>Mus musculus</i> Hsd:Athymic Nude-Foxn1 ^{nu}	Harlan Laboratories	
<i>Mus musculus</i> NOD.Cg-Prkdcscid Il2rgtm1Wjl/SzJ (NSG)	Jackson Laboratories	
Oligonucleotides		
for Primers see Table S6	This study	N/A
HTS Screen sequencing primer: STAG2 Seq Forward: AATGATACGGCGACCACCGAGATCTACACTCTTCCCTACACGACGCTCTTCCGATCTNNNNNNNNGAATTTCTCTACAAAAGCTCTTCA	This study	N/A
HTS Screen sequencing primer: STAG2 Seq Reverse: CAAGCAGAAGACGGCATACGAGATNNNNNNN NNNTTCATCATTCCTCCTCCTC	This study	N/A
HTS Screen sequencing primer: TP53 Seq Forward: AATGATACGGCGACCACCGAGATCTACACTCTTCCCTACACGACGCTCTTCCGATCTNNNNNNNNGAACTACTTCTGAAAACAACGT	This study	N/A
HTS Screen sequencing primer: TP53 Seq Reverse: CAAGCAGAAGACGGCATACGAGATNNNNNNN NNGCTTCATCTGGACCTGGGTC	This study	N/A
ON-TARGETplus Human UPF1 siRNA Smartpool 10nmol	Horizon Discovery	Cat #L-011763-00-0010
ON-TARGETplus Human mTOR siRNA Smartpool 10nmol	Horizon Discovery	Cat #L-003008-00-0010
ON-TARGETplus Human ATM siRNA Smartpool 10nmol	Horizon Discovery	Cat #L-003201-00-0010
ON-TARGETplus Human ATR siRNA Smartpool 10nmol	Horizon Discovery	Cat #L-003202-00-0010
SMARTpool: ON-TARGETplus SMG1 siRNA	Horizon Discovery	Cat #L-005033-00-0020
SMARTpool: ON-TARGETplus PRKDC siRNA	Horizon Discovery	Cat #L-005030-00-0010
SMARTpool: ON-TARGETplus PIK3CA siRNA	Horizon Discovery	Cat #L-003018-00-0010
ON-TARGETplus Non-targeting Pool	Horizon Discovery	Cat #D-001810-10
Phage display library	GeneArt	N/A
Recombinant DNA		
N/A		
Software and algorithms		
HISAT2 (version 2.0.5)	Kim et. al.[82]	N/A
StringTie (version 1.3.3)	Pertea et. al.[83]	N/A
Ballgown (version 2.6.0)	Pertea et. al.[83]	N/A
R (version 4.0.3)	R Core Team 2022	N/A
ggplot (version 3.4.1)	https://github.com/tidyverse/ggplot2/blob/HEAD/R/ggplot2-package.R	N/A
Pipeline for analyzing mutant and normal transcript abundance	MSSQL was used for initial data processing	N/A
Other		

Resource availability

Contact for Reagent and Resource Sharing

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Nicolas Wyhs (wyhs@jhmi.edu). Isogenic knockout cell lines are available through The Genetic Resources Core Facility at Johns Hopkins School of Medicine (jhbiorbank@jhmi.edu, Maryland, USA). KVS0001 generated in this study will be made available on request if available, but availability may be limited and we may require a payment and/or a completed materials transfer agreement if there is potential for commercial application.

Data availability

- Whole transcriptome RNA-seq raw and processed data have been deposited at GEO and are publicly available as of the date of publication. Accession numbers are listed in the Key Resource Table.
- Full western blot images have been deposited at Mendeley (Mendeley Data, V1, doi: 10.17632/n5s2z393z6.1) and are publicly available as of the date of publication.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

Experimental models and subject details

Cell lines

NCI-H716, NCI-H2228, HEK293, NCI-H358, RPE1, MCF10a, RPTec, LLC, RENCA, B16-F10 and LS180 cells were purchased from The American Type Culture Collection (Virginia, USA). RPE1 STAG2 2, RPE1 STAG2 5, RPE1 STAG2 6, RPE1 STAG2 8, RPTec STAG2 3, RPTec STAG2 943, RPTec STAG2 943A, RPTec STAG2 944, RPTec STAG2 945, RPTec STAG2 946, RPTec STAG2 947, RPTec STAG2 951, RPTec STAG2 952, RPTec STAG2 953, RPE1 TP53 221, RPE1 TP53 223, RPE1 TP53 224, RPTec TP53 54, RPTec TP53 588, RPTec TP53 544 isogenic knockout cell lines were generated and grown as previously described (Cook et al, iScience 2022)[33]. NCI-H358, NCI-H2228, NCI-H716, RENCA and RPE1 cells were grown in RPMI 1640 Medium (Gibco, California, USA, Cat #11875-119) supplemented with 10% Fetal Bovine Serum (FBS) (HyClone, Utah, USA, Cat #16777-006). LS180 cells were grown in EMEM (ATCC, Virginia, USA, Cat #30-2003) supplemented with 10% FBS. LLC, B16-F10 and HEK293 was grown in DMEM (Gibco, USA, Cat #11995065) supplemented with 10% FBS. RPTec cells were grown in EPITHELIAL CELL MEDIUM-Complete Kit (Science Cell Research, California, USA, Cat #4101). MCF10a cells were grown in MEGM Mammary Epithelial Cell Growth Medium BulletKit (Lonza, USA, Cat #CC3150). *In vitro*, all cells were grown at 37°C with 5% CO₂. Mycoplasma testing was performed by The Genetic Resources Core Facility at Johns Hopkins School of Medicine (Maryland, USA).

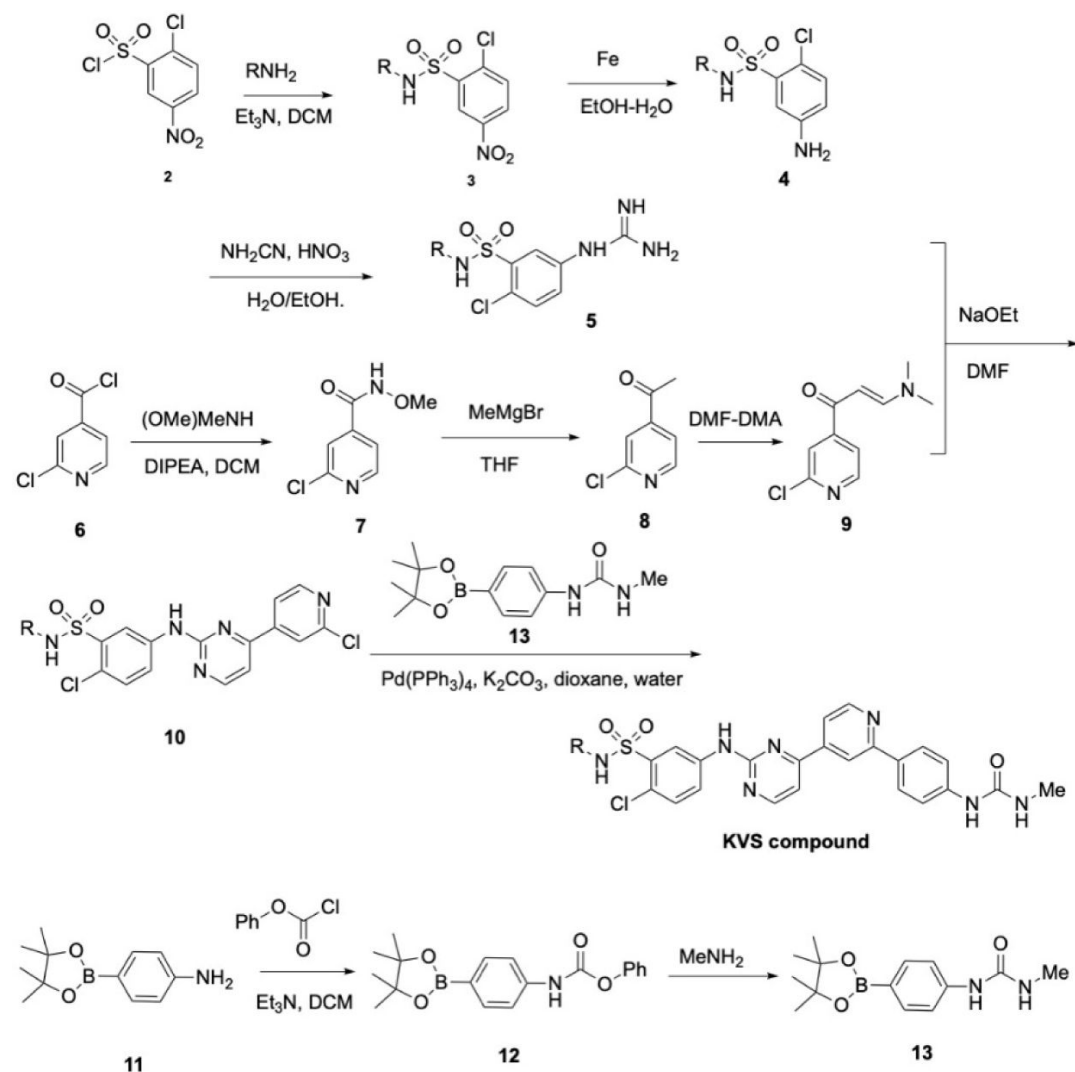
Quantitative real-time PCR (qPCR)

RNA was obtained from cells using a Qiagen RNeasy Kit (Qiagen, Maryland, USA, Cat #74104) per the manufacturer's instruction. Reverse transcription of RNA to cDNA was performed using a High-Capacity cDNA Reverse Transcription Kit (Applied Biosystems, USA, Cat #4368814) per manufacturer's instructions. Unless otherwise specified, qPCR reactions were set up using SsoAdvanced Universal SYBR Green Supermix (Bio-Rad, USA, Cat #1725270) following manufacturer's instructions and performed on a QuantStudio 3 (Applied Biosystems, USA) with manufacturer's recommended plates and plate covers. PCR thermocycling conditions were as follows: 2 min at 50°C, 2 min at 95°C, 35 cycles of 10 sec at 95°C, 10 sec at 60°C, 30 sec at 72°C. Finally a melt curve was performed starting at 50°C and ending at 95°C with 5-second incubations for imaging. For TP53 β analysis, qPCR reactions were set up using PRIMETIME™ Gene Expression

master mix (Integrated DNA Technologies, USA, Cat #1055770) and a probe with FAM reporter and TAMRA quencher (Integrated DNA Technologies, USA) following the manufacturer's instructions and concentrations. PCR thermocycling conditions were as follows: 3 min at 50°C, 10min at 95°C, 40 cycles of 10 sec at 95°C, 30 sec at 60°C. Primers used for all qPCR can be found in Table S6.

Small molecule compounds

SMG1 specific inhibitor SMG1i11 (11j)[32] and KVS0001 were synthesized by Ascendex LLC (Pennsylvania, USA). The synthesis scheme for KVS0001 is located below. Emetine was obtained from Sigma Aldrich (Cat #7083-71-8). All other small molecule hits from the screen were purchased from Selleckchem (Texas, USA).



NMD screen

We screened the Selleckchem Bioactive Compound library, a 2,658 compound library (Selleckchem, Texas, USA, Cat #L1700). The library was distributed over 33 x 96-well tissue culture plates with each plate containing library compounds, 2 NMD positive controls (emetine, 12 $\mu\text{g/mL}$) and 6 NMD negative controls (DMSO). We mixed the three cell lines, RPE1 STAG2 2, RPE1 STAG2 8, and RPE1 TP53 221, in equal ratio and dosed with the compound library at 10 μM for 16 hours. Cell pools were harvested by washing plates 2x with PBS and frozen. RNA extraction and reverse transcriptase cDNA conversion were performed as described in *Targeted RNA-seq*. Quality control was performed by qPCR of one NMD positive and one NMD negative control well from each plate

to confirm that increased expression of the positive control (emetine) wells was observed. All samples were prepared for sequencing by amplifying cDNA using primers containing both a plate (forward primer) and well (reverse primer) barcode which amplifies the knockout cell line specific mutation for STAG2 and TP53 (See Table S6). Primers were obtained from Integrated DNA Technologies (Iowa, USA). PCR was performed using Phusion Flash High-Fidelity PCR Master Mix (Thermo Fisher, USA, Cat #F548S) for 1 min at 98 °C, 29 cycles of 10 sec at 98 °C, 15 sec at 64 °C, 15 sec at 72 °C, and 5 min at 72 °C. Samples were then well barcoded using a similar PCR setup for 2-4 cycles. Samples were then pooled, cleaned up with AMPure beads (Beckman Coulter, California, USA, Cat #A63880), and sequenced on an Illumina HiSeq2500 using manufacturer's instructions (150 cycle single read) for an average of 56,753 reads per well. The screen was scored by calculating the number of sequencing reads matching each mutant and reference transcript and calculating a Mutant Allele Fraction (MAF) correcting for the number of cell lines and heterozygous mutations.

NMD positive and negative control data were pooled, respectively, and averaged across all plates to determine the hit threshold of $MAF > 0.46$. This value, 5 standard deviations above the mean for the DMSO controls, was chosen to ensure no false positives in the DMSO controls were observed. A compound was considered a hit if the MAF for all three cell lines was greater than this value. All results from the screen are reported as MAF.

Targeted RNA-seq

RNA extraction of *in vitro* cells was performed using BioRad SingleShot Cell Lysis kit (Bio-Rad, California, USA, Cat #1725080) scaled down to 50 µl reactions per well of 96-well plate per manufacturer's instructions. Tissue culture cells were lysed directly on the tissue culture plate. For RNA extraction of *in vivo* studies, tissues were harvested and placed in RNA Later (Invitrogen, Maryland, USA, Cat # AM7020) and stored at -80°C until RNA extraction. RNA extraction of tissues was performed using a Qiagen RNeasy Kit (Qiagen, Maryland, USA, Cat #74104) per manufacturer's instruction with tissue homogenization in 600 µl of RLT buffer via dounce homogenizer followed by QIAshredder (Qiagen, Maryland, USA, Cat #79656). All RNA quality was assessed by Agilent TapeStation 2200 (Agilent, California, USA, Cat #G2964AA) and the Agilent RNA ScreenTape (Agilent, California, USA, Cat #5067-5576) with Agilent RNA ScreenTape Sample Buffer and Ladder (Agilent, California, USA, Cat #5067-5577, Cat #5067-5578) per manufacturer's instruction. Reverse transcription to cDNA was performed using High-Capacity cDNA Reverse Transcription Kit (Applied Biosystems, USA, Cat #4368814) per the manufacturer's instructions. cDNA was amplified using cDNA-specific primers with at least one primer (forward or reverse) covering an exon-exon boundary (See Table S6 for primer sequences). Primers were obtained from Integrated DNA Technologies (Iowa, USA). PCR was performed using Phusion Flash High-Fidelity PCR Master Mix (Thermo Fisher, USA, Cat #F548S) for 1 min at 98 °C, 30 cycles of 10 sec at 98 °C, 15 sec at 64 °C, 15 sec at 72 °C, and 5 min at 72 °C. Sequencing libraries were prepped from samples by addition of well barcodes using the same method described above for an additional 2-6 cycles. Libraries were pooled, cleaned up with AMPure beads (Beckman Coulter, California, USA, Cat #A63880) and sequenced on an Illumina Miseq using manufacturer's instructions (150 cycle single read).

Mutant Allele Fraction (MAF) analysis

Mutant allele fraction was determined by processing fastq files using HISAT2 (version 2.0.5) and aligning to a pseudo reference genome consisting of only the mutant or wild-type amplicon sequences for targeted regions. The mutant allele fraction was determined by taking a ratio of the number of mutant transcripts to the total number of transcripts from the region in question. Initial data processing was performed in MSSQL and Excel.

Whole transcriptome RNA-seq

LS180 or NCI-H358 cells were run in biological duplicate, treated with DMSO or 5 μ M LY3023414. For RNA extraction, cells were pelleted, frozen in liquid nitrogen, and stored at -80°C until RNA extraction. RNA extraction was performed using a Qiagen AllPrep DNA/RNA Mini Kit (Qiagen, Maryland, USA, Cat# 80204) per manufacturer's instruction with cell homogenization and lysis in RLT buffer with a QIAshredder (Qiagen, Maryland, USA, Cat# 79656). RNA quality control using Agilent Tapestation 2200 (Agilent, California, USA, Cat# G2964AA) and the Agilent RNA ScreenTape (Agilent, California, USA, Cat# 5067-5576) with Agilent RNA ScreenTape Sample Buffer and Ladder (Agilent, California, USA, Cat# 5067-5577, Cat# 5067-5578) per manufacturer's instruction. Library prep using Illumina RNA library prep kit (Illumina, California, USA, Cat #RS-122-2001) and sequenced on an Illumina HiSeq 4000 150 cycle paired-end using manufacturer's instructions.

RNA-seq analysis

Sequencing reads aligned to Hg38 using HISAT2 (version 2.0.5), RNA alignment metrics using CollectRnaSeqMetrics (Picard, version 2.20.2). Exon skipping was determined using IGV Viewer Sashimi Plots[84]. The average number of bases sequenced per sample and percent aligned in LS180 was 5.14e9 bases (range 5.07e9-5.20e9) and 77.8% (range 76.1%-78.7%) and for NCI-H358 was 5.58e9 bases (5.19e9-5.87e9) and 79.4% (range 77.2%-82.7%). Mean allele frequency was determined using VarScan 2[85] by generating the ratio of Read 1 (mutant) to read2 (wild type) transcripts. Mutations were only considered if they were heterozygous, and contained at least 5 reads at the somatic mutation or indel site in all 4 samples being compared (biological duplicates of treated and untreated).

Compound response curves

We performed a 6-point dose-response curve by treating cell pools for 14 hours with single compounds. Cell pools consisted of three isogenic knockout cell lines grouped by parental cell line with readout via targeted sequencing RNA (see *Targeted RNA-seq* for details). The effect of the compound was determined by calculating the MAF by comparing the abundance of the expected mutation and compared to the wild type within each isogenic pool (see *Mutant Allele Fraction (MAF) analysis* for details).

Immunohistochemistry (IHC)

IHC was performed on cell lines as previously described[86].

Western blots

Cells were lysed using RIPA buffer (Thermo Fisher, USA, Cat #89901) containing 1x protease inhibitor (Thermo Fisher, USA, Cat #4693159001) on ice for 30 minutes. Samples were then centrifuged at max speed for 3 minutes at 4°C in a QIA shredder (Qiagen, Maryland, USA, Cat #79654) before being transferred to a new sample collection tube. Protein was quantified using a BCA assay (Thermo Fisher, USA, Cat #23227) per the manufacture's instructions. Gels were run by loading 50 μ g of total protein per sample into 15 well polyacrylamide gels (Bio-Rad, California, USA, Cat #456-1086) and run for 30 minutes at 200V. Gels were then transferred using the manufacturer's instructions (based on size) to nitrocellulose membrane using a Bio-Rad turbo transfer apparatus (Bio-Rad, USA, #170-4270).

Membranes were blocked for 1 hour with 3% milk-TBS-Tween before being incubated overnight in primary antibody (concentration dependent on antibody). Primary antibodies and concentrations can be found in the key resources table. Membranes were washed 4 times for 5 minutes each with TBS-Tween. Secondary antibody was applied at 1:2500 using either α -rabbit (Jackson

ImmunoResearch, Pennsylvania, USA, Cat #111-035-006) or α -mouse (Jackson ImmunoResearch, Pennsylvania, USA, Cat #115-035-006). Membranes were imaged using Pierce™ ECL Western Blotting Substrate (Thermo Fisher, USA, Cat #32106) following the manufacturer's instructions on a Bio-Rad Chemidoc (Bio-Rad, California, USA). Phospho-UPF1 westerns were performed as detailed above with the following exceptions: secondary antibodies were used at 1:10,000 either α -rabbit (Licor Biosciences, USA, Cat #926-32213) or α -goat (Licor Biosciences, USA, Cat #926-68074). Membranes were imaged using an Odyssey CLx (Licor Biosciences, USA) following the manufacturer's instructions.

siRNA

We performed knockdown of kinase proteins using siRNA. LS180 and NCI-H358 cells were plated in 96-well plates (Costar, USA, Cat #3595) at 5,000 cells per well. Cells were transfected with DharmaFECT1 transfection reagent (Horizon, USA, Cat #T-2001-02) and either 50 nM control or kinase-specific pooled siRNA. siRNA used in this experiment can be found in the key resources table. Knockdown efficacy was confirmed using qPCR (see *Quantitative real-time PCR (qPCR)* for details). NMD target gene mutation transcript levels were determined using next-generation sequencing and MAF (see *Targeted RNA-seq* for details).

Bispecific scFv construction

We utilized a bispecific antibody against CD-3 and TP53 wild-type peptide 'TYSPALNKM' (residues 125-134) presented in a HLA-A24 MHC-I molecule scFv. This bispecific antibody was identified by panning a phage display library. The scFv-bearing phage library was constructed similarly as described in detail previously with some modifications[87]. Briefly, oligonucleotides were synthesized by GeneArt (Thermo Fisher, USA) using trinucleotide mutagenesis (TRIM) technology to diversify complementarity-determining region (CDR)-L2, CDR-L3, CDR-H1, CDR-H2, and CDR-H3. A FLAG (DYKDDDDK) epitope tag was placed immediately downstream of the scFv, which was followed in frame by the full-length M13 pIII coat protein sequence. The total number unique clones obtained was determined to be 3.6×10^{10} . Panning details can be found in the reference section[86, 87].

Kinase target assay

Kinase kinativ experiments were performed by ActivX Biosciences (San Diego, USA)[54].

Quantitative presentation of HLA-bound neoantigens via HPLC-Mass Spectrometry

Identification and quantitation of HLA-presented neoantigens was performed as previously described by Complete Omics Inc.[57]. Briefly, cells were treated *in vitro* for 24 hours with either DMSO or 5 μ M KVS0001. Cells were cross-linked and immunoprecipitated with pan-HLA antibodies to obtain cell surface MHC-presented peptides. Mass spectrometry was performed in the presence of heavy labeled peptide to serve as an internal (loading) control to quantify the presence of the presented neoantigen. The presented peptides were identified in a preliminary MS screen of 187 candidate peptides predicted by using the union of NetMHC and Predictor of Immunogenic Epitopes (PRIME) predictions[58, 59].

CD3-TP53scFv bispecific co-culture

Co-culture of bispecific antibody was performed using volunteer human donor T-Cells. T-cell enrichment and activation was performed as previously described [88]. Briefly, PBMCs are incubated with OKT3 antibody (Biolegend) for 3 days. T-cells were then expanded in RPMI-1640 with 10% FBS and 1% penicillin-streptomycin, recombinant IL-2 (Proleukin, Prometheus Laboratories) and IL-7 (BioLegend) for at least 15 days before use. NCI-H716 cells were labeled

with HLA-A24 using retrovirus as previously described[86]. Briefly, the MSCV retroviral expression system (Clontech, USA, Cat #634401) was used to overexpress HLA-A*24-T2A-GFP in target cells. Expression was confirmed by flow cytometry. NCI-H2228 cells were not labeled as they express this specific HLA endogenously. For NCI-H716_A24 and NCI-H2228, 40,000 cells were plated in 96-well plates, and co-incubated with 40,000 activated T-cells, and 12.5 pg/mL of TP53-CD3 bispecific antibody. Cells were then dosed with 1.25 μ M of KVS0001, SMG1 specific inhibitor or 200 mg/mL of 5-fluoruracil (Sigma Aldrich, USA, Cat #F6627) for 24 hours. Readout of interferon (IFN- γ) was performed by ELISA following the manufacturer's instructions (R&D Systems, USA, Cat #SIF50C). Cell killing was assayed by luciferase levels following the manufacturer's instructions (Promega, USA, Cat #E1501) and read out on a Synergy H1 Microplate reader (BioTek, USA). NCI-H716_A24 cells were labeled with luciferase via lentiviral transduction following the manufacturer's recommendations (OriGene, USA, Cat #PS100071).

Animal protocols

Animal research was approved and overseen by Johns Hopkins University Institutional Animal Care and Use Committee (ACUC) approved research protocol M018M79. Mice are housed in individually ventilated caging (Allentown, New Jersey, USA) at a maximum 5 animals per cage. Cages are changed every 14 days. Enrichment is provided through paper bedding, paper hut, and some food placed in the bottom of the cage. Facility is maintained between 70 and 72 degrees Fahrenheit on a 12-hour light-dark cycle. Mice standard diet is ad lib Teklad Global 18% Protein Extruded Rodent Diet, autoclaved (Envigo, Huntingdon, UK) and acidified water via sipper tube.

In Vivo tumor models

Cells for tumor inoculation in each mouse were grown to 70-80% confluency *in vitro*. Cells were harvested with trypsin (Gibco, California, USA, Cat #25300054) and suspended in either PBS for LS180 cells or 50% PBS, 50% Matrigel Phenol Red Free Standard Formulation (Corning, New York, USA, Cat #356237) for NCI-H358 cells. For both cell lines 1e6 cells were placed subcutaneously and grown to approximately 200 mm³. Animals were randomized into treatment groups prior to treatment by tumor size. All tumor volumes were measured via caliper twice weekly. Mice with tumors <150 mm³ at 11 days post inoculation were excluded from experiments.

Human xenograft experiments

6 to 8 weeks old *Mus musculus* Hsd:Athymic Nude-Foxn1^{nu} mice (referred to as nude mice) were purchased from Harlan Laboratories (Indiana, USA). Only female mice were used as gender was not considered to be a significant confounder in the experiment. LS180 cells were inoculated at 1.0 x 10⁶ cells per mouse in the left mouse flank or NCI-H358 cells were inoculated at 7.5 x 10⁵ cells per mouse in the right flank. For single-dose experiments, mice were orally dosed via gavage at 0, 40, or 60 mg/kg of LY3023414 (Selleckchem, Texas, USA, Cat #S8322) in 1% Methylcellulose (Sigma Aldrich, Missouri, USA, Cat #M6385). KVS0001 was dosed at 30 mg/mL intraperitoneal (IP) in 0.5% Dimethyl sulfoxide (DMSO)(Sigma Aldrich, USA, Cat #C6295), 10% cremaphor (Sigma Aldrich, USA, Cat #C5135) and 2% glycerol (Sigma Aldrich, USA, Cat #G5516). Mice were given physical exams prior to euthanasia at designated endpoints according to the approved research protocol. Tumors, spleen, blood, and lungs were harvested post-euthanasia for further analysis. Tumors were harvested at indicated times and stored in RNA Later (Invitrogen, Maryland, USA, Cat # AM7020) at -80 for further analysis.

Syngeneic mouse tumor experiments

6 to 8 weeks old immune compromised *Mus musculus* BALB/cAnNCrI (referred to as BALB/c) and C57BL/6NCrI (referred to as C57BL/6N) mice were purchased from Charles River GmbH (Germany). Only female mice were used as gender was not considered to be a significant confounder in the experiment. Murine cancer cell lines were placed in the left mammary fat pad of female mice at

the following densities: LLC 1.0×10^6 , RENCA 1.0×10^6 , and B16-F10 0.2×10^6 . Mice were randomized around day 7 post-implantation when tumor sizes had reached approximately 30–40 mm³. Mice were dosed with 30 mg/kg KVS0001 or vehicle control via IP injection alternating the right and left side daily for 28 days (Figure 5C). Mice were given physical exams before euthanasia at designated endpoints according to the approved research protocol.

In Vivo TP53 bispecific experiments

For the NCI-H716 *in vivo* bispecific experiments 2.5×10^6 NCI-H716_A24 expressing cells were placed orthotopically by IP injection into NOD.Cg-Prkdcscid Il2rgtm1Wjl/SzJ (referred to as NSG) mice (Jackson Laboratories, USA). On day 2 mice were randomized to ensure a balanced tumor burden in all groups. Luminescence was measured by injecting mice with 150 µL of RediJect D-Luciferin Ultra Bioluminescent Substrate (PerkinElmer, USA, Cat #770505), and anesthetized using isoflurane in an induction chamber for 5 minutes. Readouts and analysis were performed on an IVIS Spectrum imaging system and Living Image software (Perkin Elmer, USA).

Data reporting

No statistical methods were used to predetermine the sample size. Statistical analysis was performed using R and Excel. All animal experiments were randomized being sure to alter the first cage dosed and location of cages with a row of the rack. Randomization was performed by animal tumor size, unless otherwise indicated. The investigators responsible for weight and tumor measurements were blinded to the allocation, treatment, and outcome assessment of experiments. The investigators processing mouse tissue processing were blinded to the allocation, treatment, and outcome assessment of experiments. The investigator harvesting the tissues was aware of the allocation, treatment, and outcome assessment of experiments.

Statistical Testing

Chi-Squared testing was performed using counts of cancer cell line mutations predicted to undergo NMD and RNA recovered and `prop.test()` in R. Mann-Whitney testing performed with `wilcox.test()` in R. Quantiles calculated utilizing z scores with `quantile()` in R. Student T-Test and one-way ANOVA with Dunnett's test and Student's t-test were performed in R and Excel.

RPE1 STAG2 Knockout Clone 2

chrX:123191737insA Exon 15 of 35	
Wild Type:	GATCC AGAGG
Alleles 1 and 2:	GATCCAAGAGG

RPE1 STAG2 Knockout Clone 8

chrX:123191735insT Exon 15 of 35	
Wild Type:	GAGAT CCAGA
Alleles 1 and 2:	GAGATTCCAGA

chrX:123179053delTGGAAG Exon 8/35	
Wild Type:	CTCAGTGAAGAA
Alleles 1 and 2:	CTCAG-----AA

RPE1 TP53 Knockout Clone 221

chr17:7579535insT Exon 4 of 11	
Wild Type:	ATTGTT CAAT
Alleles 1 and 2:	ATTGTTTCAAT

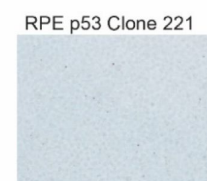
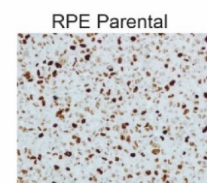
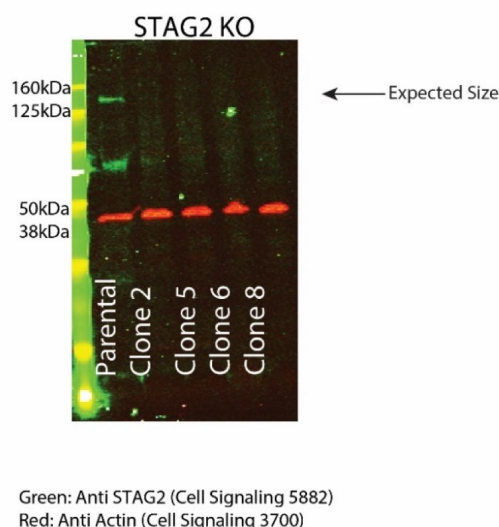


Figure S1

Next generation sequencing results depicting genomic mutations at the CRISPR target area in *STAG2* (top left) and *TP53* (top right) isogenic cell line clones in the RPE1 cell line which were used in subsequent experiments. Western blot showing *STAG2* protein loss in four independent clones of which only clones 2 and 8 were used in subsequent experiments (bottom left). Note that *STAG2* clone 8 has two independent biallelic indel events. The mutation in exon 15 was used for scoring relative transcript levels in the screen and for subsequent work. IHC results confirming p53 protein loss in RPE TP53 221 (bottom right).

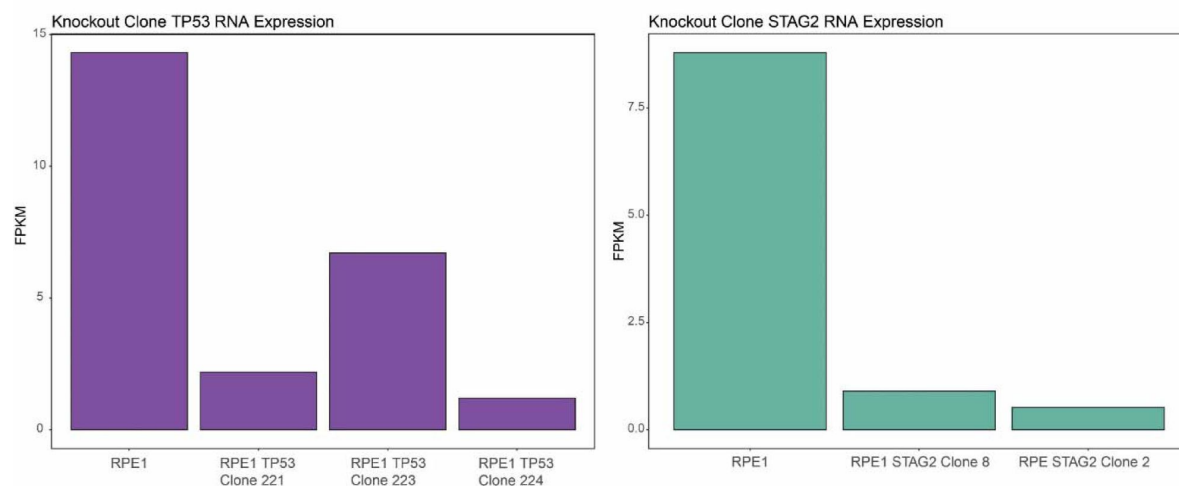


Figure S2

Fold change in RNA expression levels from whole transcriptome RNA-sequencing data for *STAG2* and *TP53* knockout clones in the RPE1 cell line background. RPE1 TP53 clone 221, RPE1 STAG2 clone 2 and RPE1 STAG2 clone 8 were used in the HTS screen. RPE1 TP53 clone 223 and RPE1 TP53 clone 224 were used in [Figure S6](#). Note clone 223 has a 9bp in-frame deletion in one allele and an out of frame deletion on the other allele, presumably accounting for the higher level of expression.

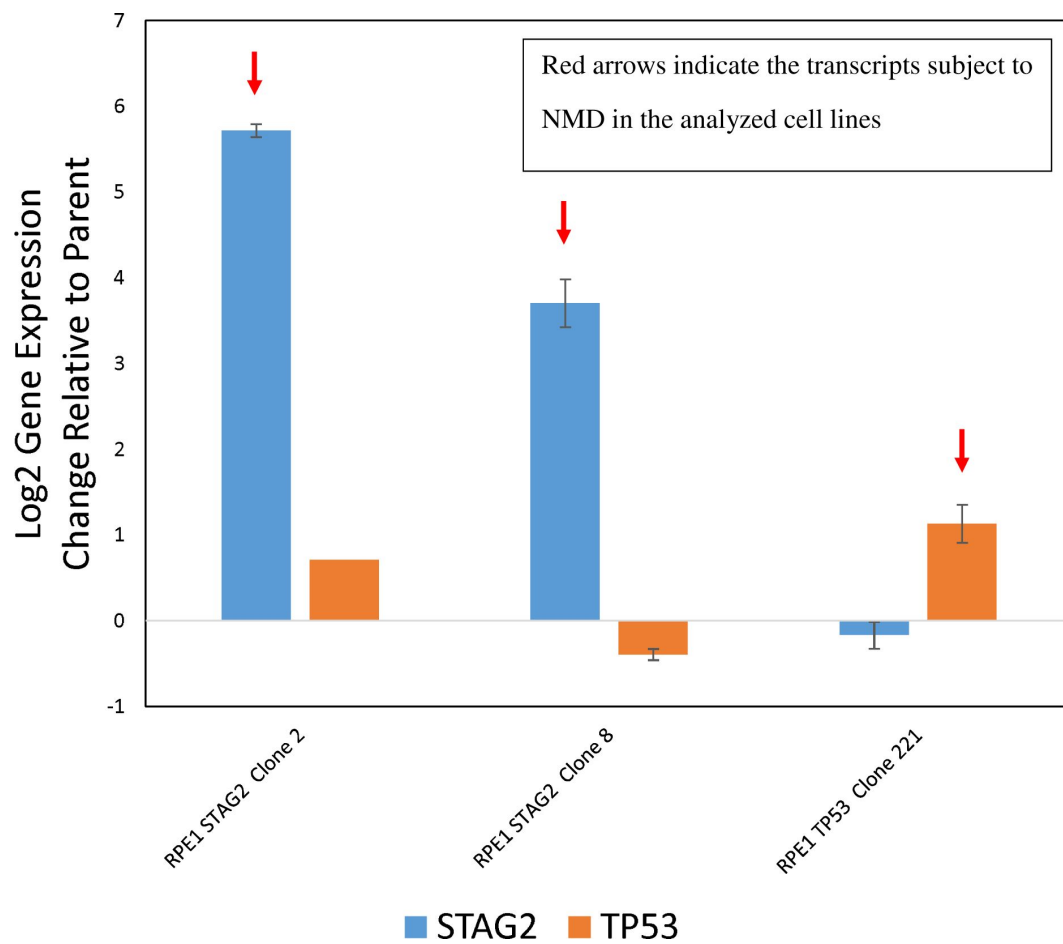


Figure S3

RNA transcript level changes based on qPCR in *STAG2* and *TP53* knockout clones treated with the known NMD inhibitor emetine at 12 mg/mL. Error bars show standard deviation of three biological replicates. All changes are statistically significant by student t-test ($p < 0.05$).

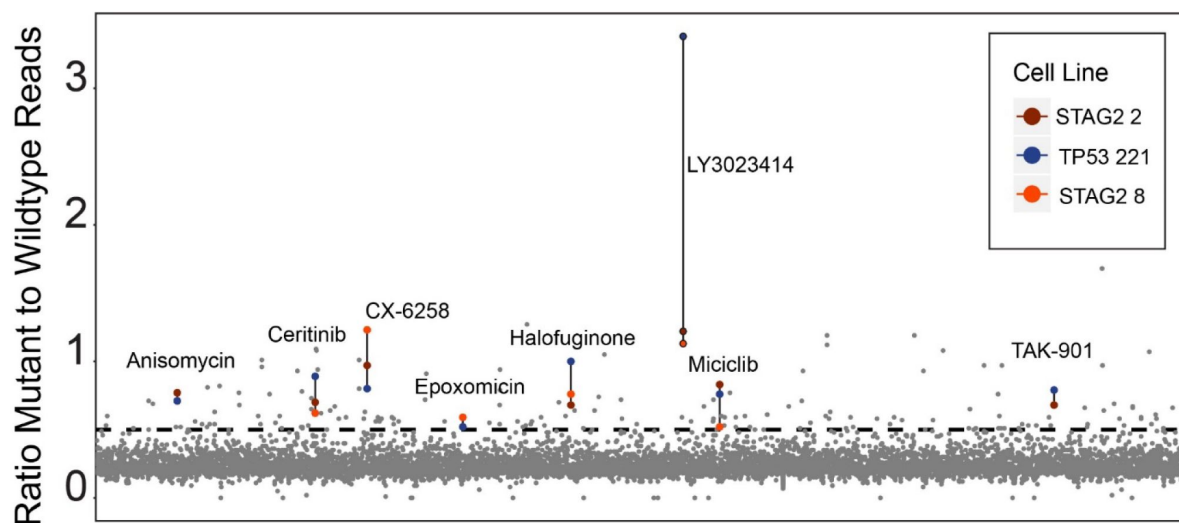


Figure S4

Primary screen results from high throughput assay. The x-axis represents all 2,658 compounds screened at 10 μ M, y-axis shows ratio of mutant to wild type reads for each of the three isogenic cell lines. Higher values indicate more mutant RNA reads, representing inhibition of NMD. The dotted line at 0.46 is the cutoff for a hit to be called (5 standard deviations above DMSO treated samples). Colored data points demarcate the eight hit compounds in the three screened cell lines.

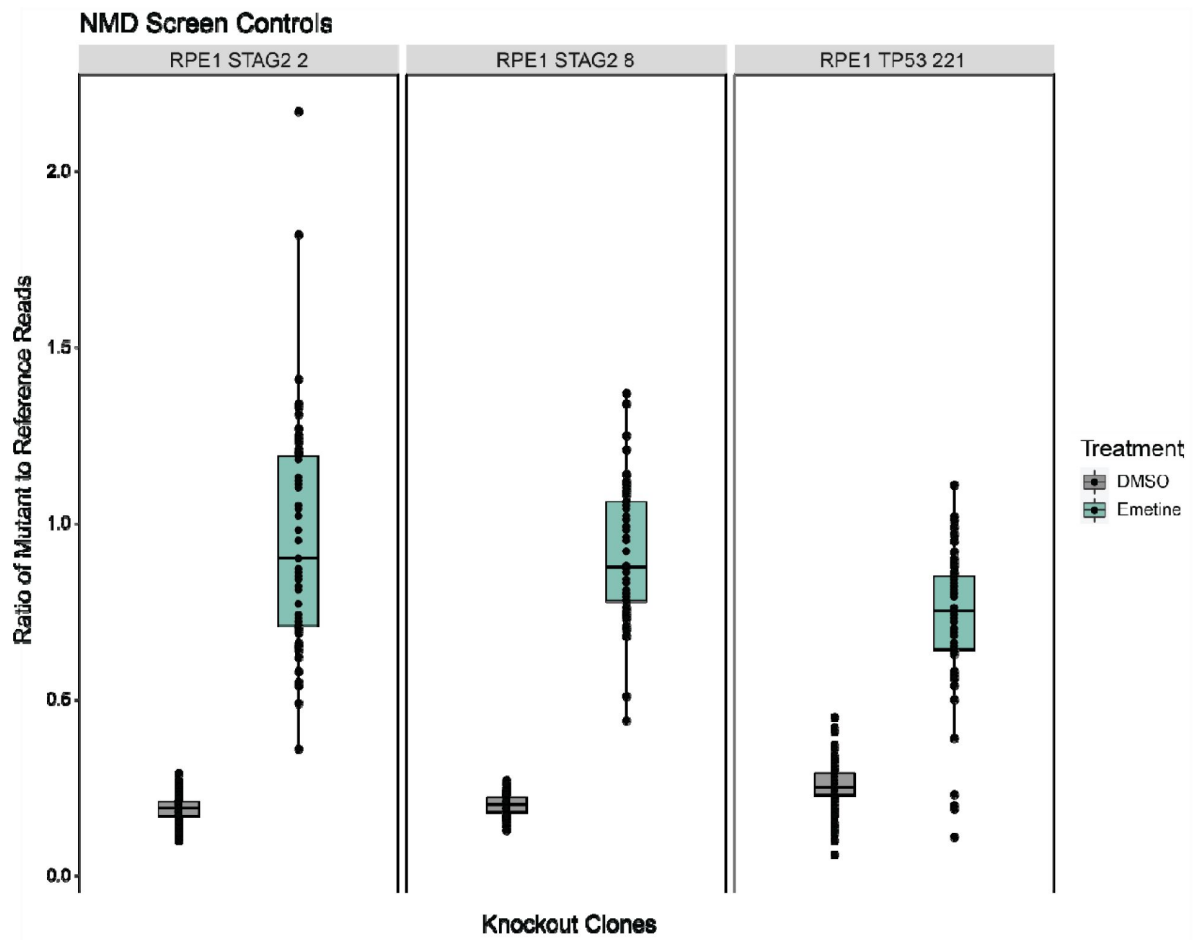


Figure S5

Emetine and DMSO control sample data from the HTS for each of the three clones used as measured by deep-targeted RNA sequencing. 6 DMSO and 2 emetine (12mg/mL) samples were included in each dosing plate for a total of 198 DMSO and 66 emetine measurements in each boxplot.

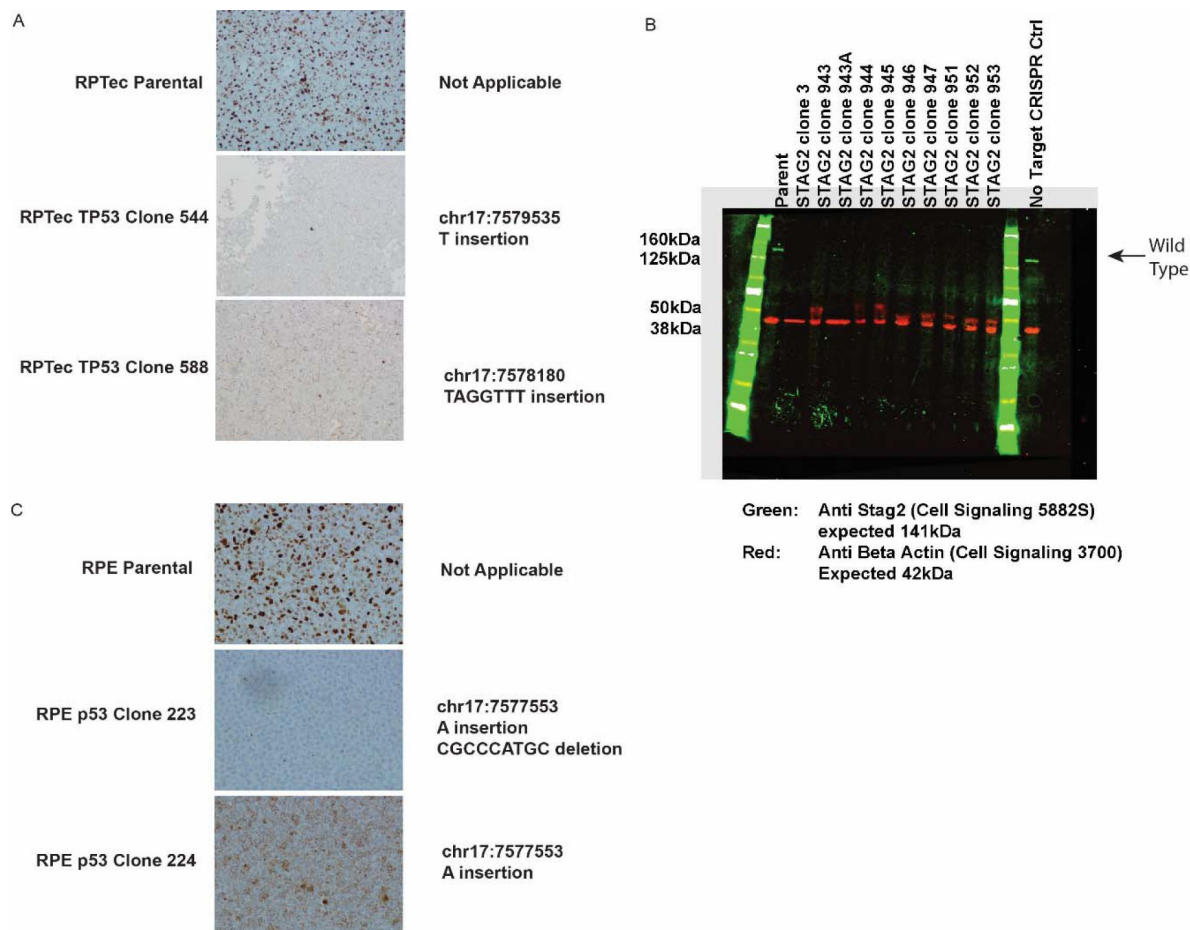
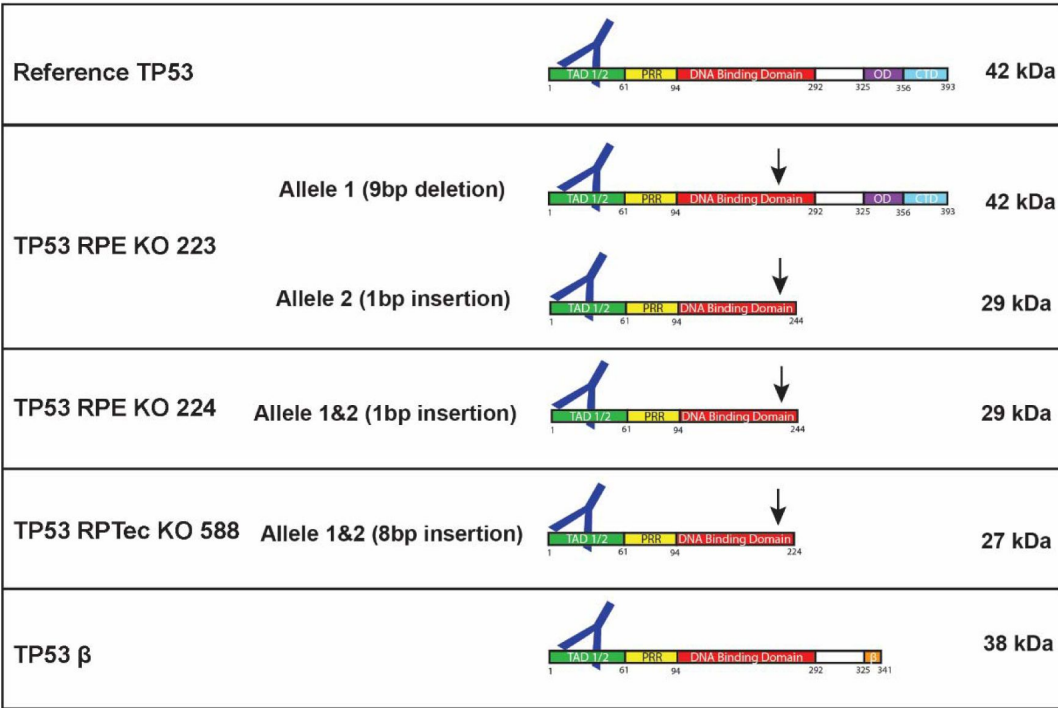


Figure S6

A) IHC staining of TP53 in RPTec *TP53* knockout clones. **B)** Western blot against STAG2 on RPTec *STAG2* knockout clones demonstrating successful knockout at the protein level of all ten clones. Out of frame indels were confirmed in all clones by NGS (data not shown). The arrow indicates the expected size for full-length STAG2 protein. **C)** IHC staining of TP53 protein on RPE1 *TP53* knockout clones. Clone 223 contains the same truncating mutation found in clone 224 on allele 1 and has a 9 bp in-frame deletion that preserves some full length TP53 α and TP53 β isoform expression on allele 2 (See **Figure 1E** [↗](#)).

Figure S7

Protein schematic cartoons showing indel mutation site and expected size of various *TP53* knockout clones used in this study. Note, RPE *TP53* 223 has an in-frame deletion in the DNA binding domain.

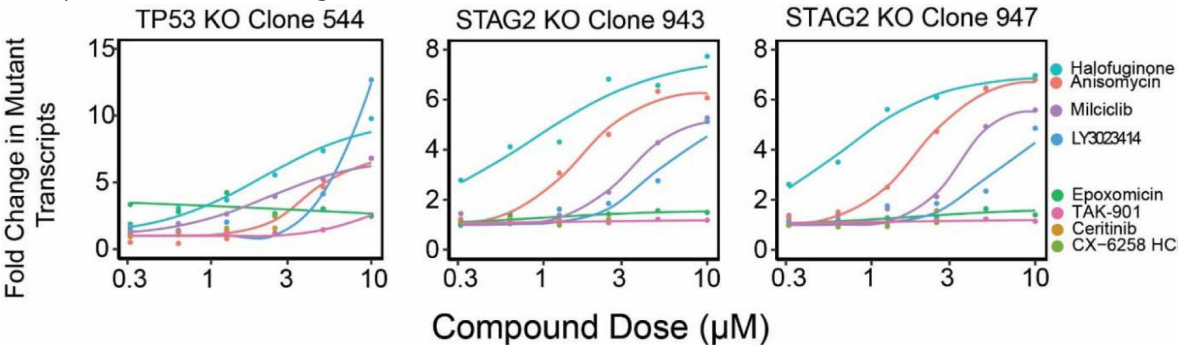


 TP53 binding antibody (Clone DO-1)

↓ Location of indel (out of frame unless otherwise indicated)

Figure S8

Fold change in mutant RNA transcription levels for *STAG2* and *TP53* in three knockout cell lines from the RPTec background containing out of frame indels targeted by NMD. Each line was treated with each of the eight hit compounds from the screen. The 10 μ M dose is also shown in [figure 1C](#).



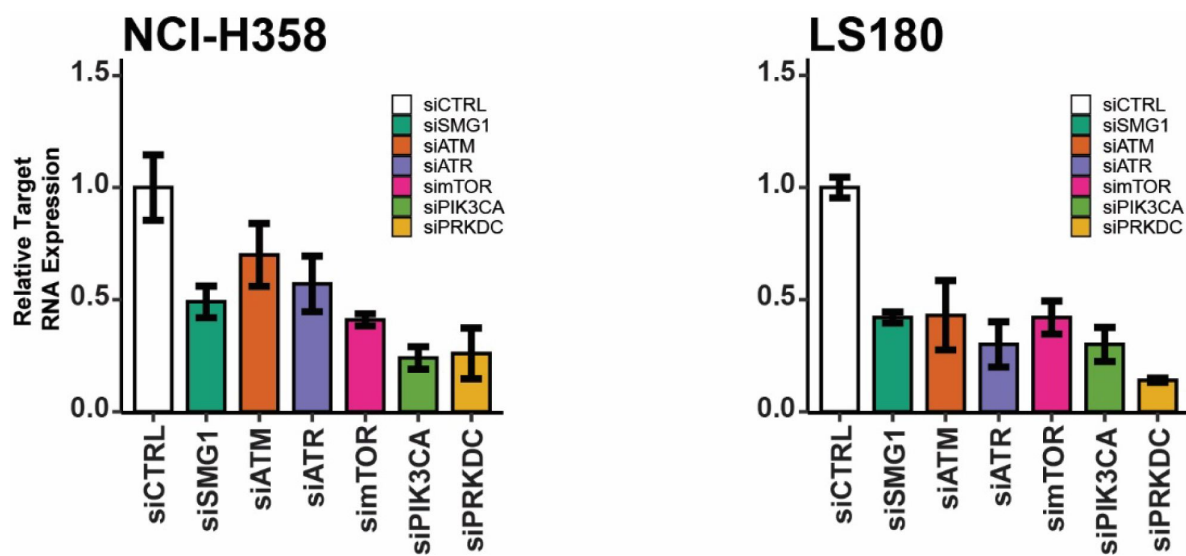


Figure S9

RNA expression levels of kinases post siRNA targeting in NCI-H358 and LS180 cells by qPCR. Cells are treated with siRNAs targeting genes known to be inhibited by LY3023414. Error bars represent 95% confidence intervals.

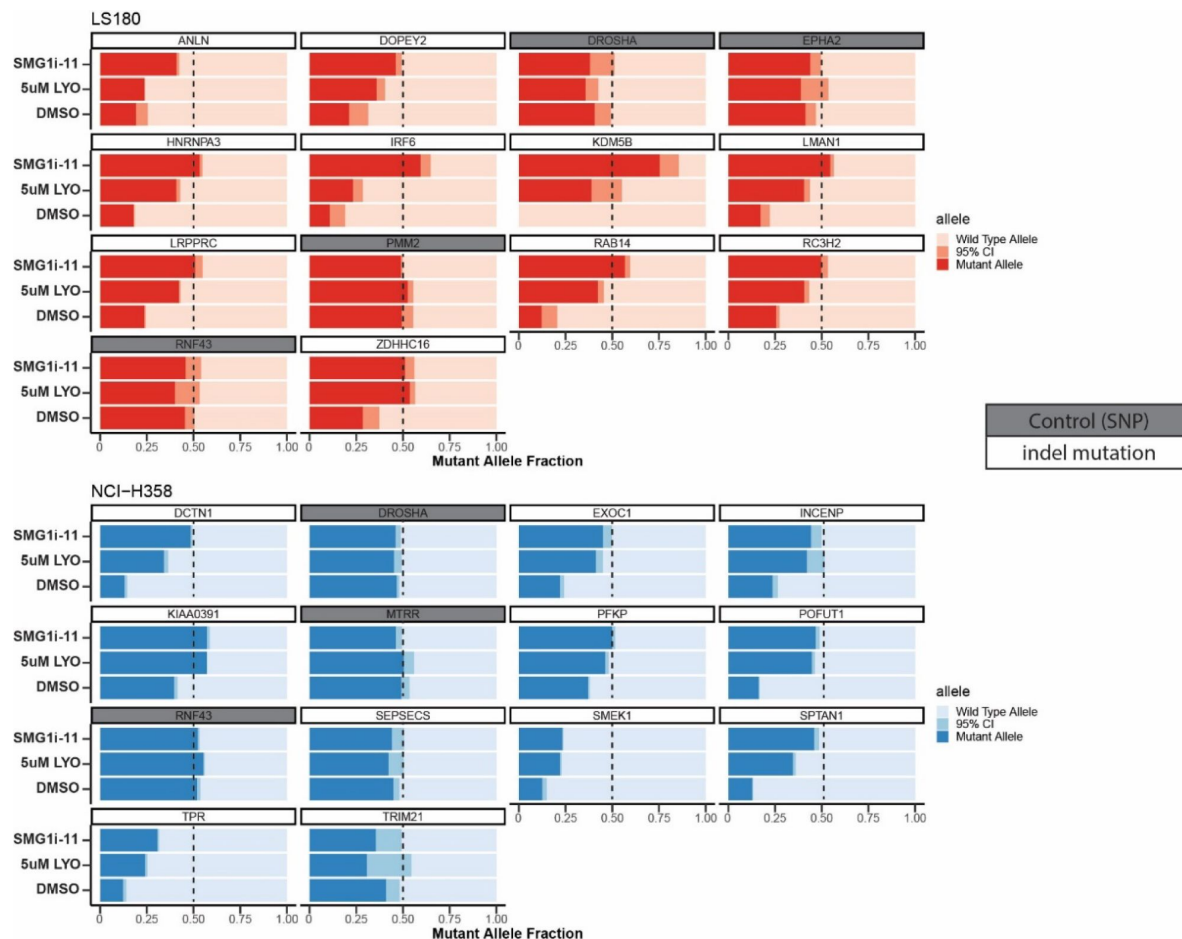


Figure S10

Slider plots showing mutant allele fraction relative to total reads measured by deep-targeted RNA-sequencing for *in vitro* treated LS180 (top in red) or NCI-H358 (bottom in blue) cells with 5 μ M LY3023414 (labeled LYO) or a previously described SMG1 inhibitor SMG1i-11 at 1 μ M. Gene names are shown in the boxes above each slider plot, genes highlighted in gray are common SNPs and serve as a negative control (not expected to change). In the case of the control SNPs, the mutant allele refers to the non-reference genome allele. *TRIM21* did not show a large change in expression with either small molecule, while *ANLN* did not show a difference with LY3023414 but did respond to SMG1i-11. The remaining genes responded to NMD inhibition by both LY3023414 and SMG1i-11.

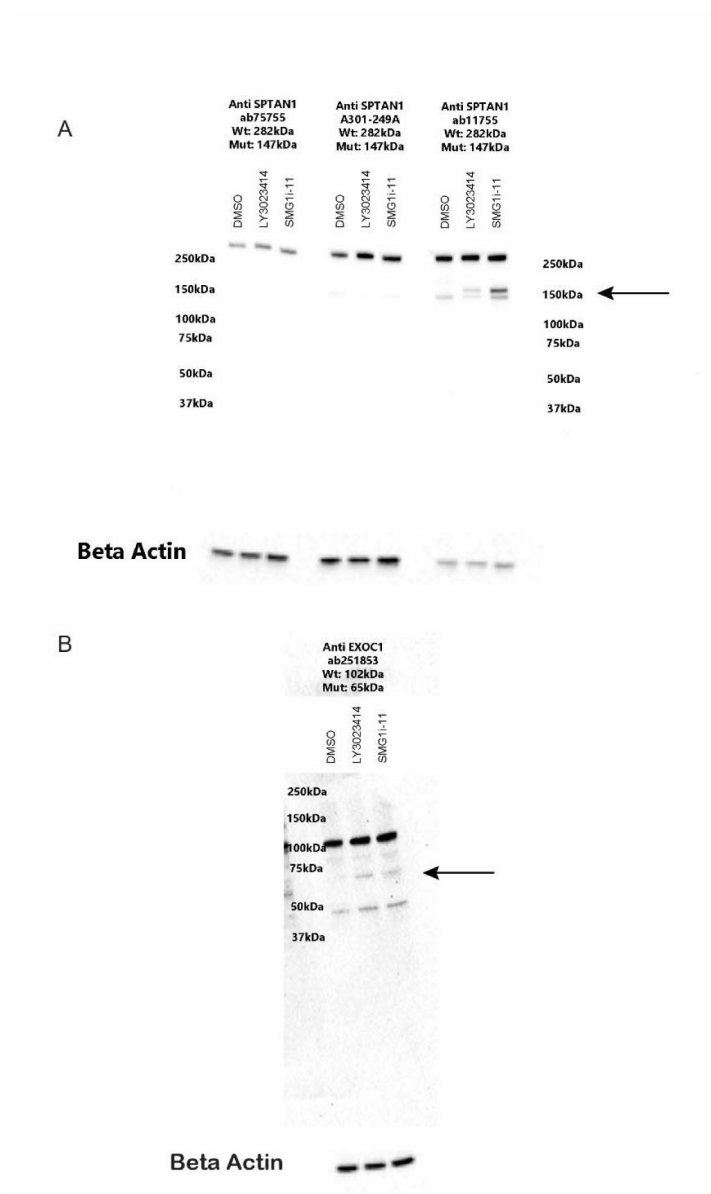
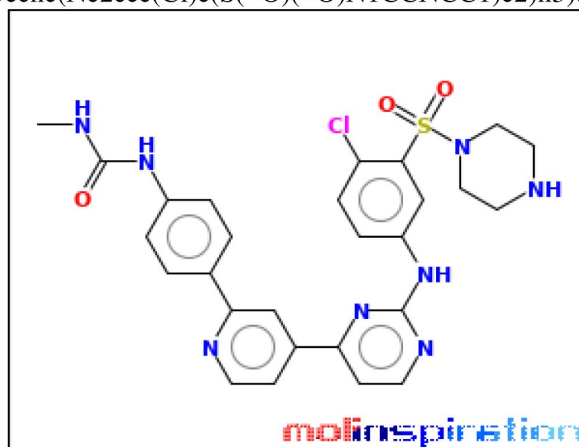


Figure S11

A) Western blot showing SPTAN1 expression after treatment with DMSO, 5 μ M LY3023414, or 1 μ M SMG1 inhibitor SMG1i-11 (lanes 1,2,3 respectively for each antibody). The arrow indicates the expected size of the mutant (NMD targeted) protein. Antibody ab75755 (Abcam) binds C-terminal to the out of frame indel and does not show mutant protein as expected. Antibody A301-249 (Bethyl) is polyclonal and also did not bind mutant protein. Antibody ab11755 (Abcam) is located N-terminal to the indel and does display mutant protein expression. **B)** Western blot showing EXOC1 expression after treatment with DMSO, 5 μ M LY3023414, or 1 μ M SMG1i-11 (lanes 1,2,3 respectively). The arrow indicates the expected size of the mutant (NMD targeted) protein.

KVS0001

CNC(=O)Nc5ccc(c4cc(c3ccnc(Nc2ccc(Cl)c(S(=O)(=O)N1CCNCC1)c2)n3)ccn4)cc5

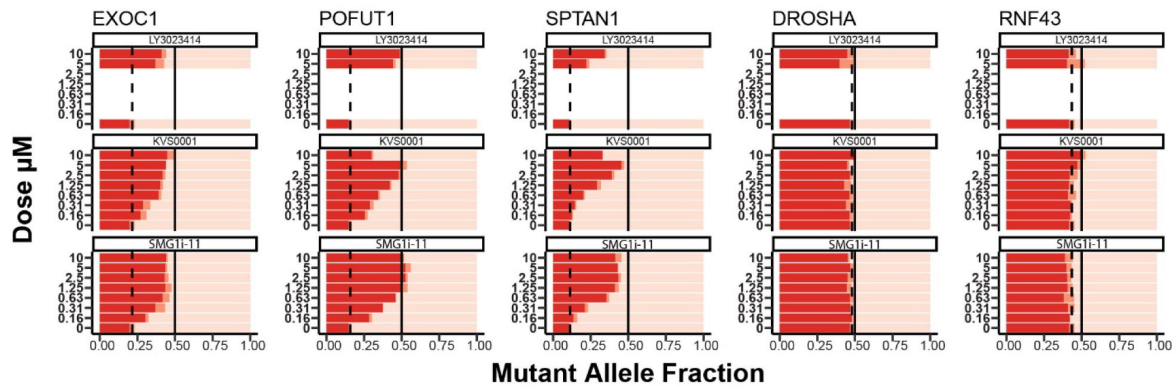


miLogP	3.32
Total Polar Surface Area (Å ²)	141.24
Number of atoms	40
Molecule Weight (Da)	579.09
Number of ON	11
Number of OHNH	4
Number of violations of rule of 5	2
Number of rotatable bonds	7
Volume (Å ³)	485.07

Figure S12

Biophysical properties for novel SMG1 inhibitor KVS0001.

A



B

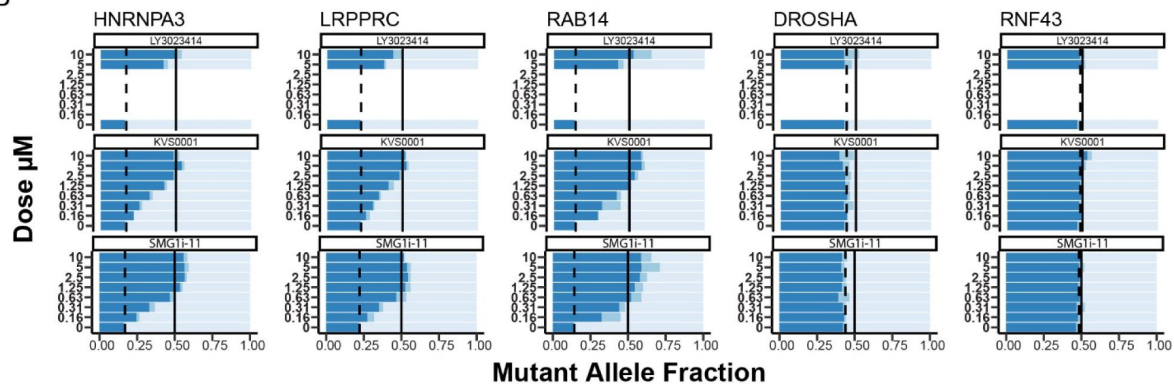


Figure S14

A) Slider plots showing mutant allele fraction measured by deep-targeted RNA-sequencing for genes from NCI-H358 (top in red) and **B)** LS180 (bottom in blue) treated in a dose response *in vitro* with novel NMD inhibitor KVS0001. *DROSHA* and *RNF43* are common heterozygous SNPs and serve as a negative control. In the case of the control SNPs, the mutant allele refers to the non-reference genome allele. Only the highest two concentrations were tested on LY3023414 which served as a positive control in this experiment. The dotted line indicates the mutant expression with DMSO treatment and the solid line is a reference for equal expression of both the wild type and mutant alleles.

Figure S15

A) Western blot showing expression of LMAN1 in LS180 cells treated with DMSO (lane 1), KVS0001 at 5 μ M (lane 2) or SMG1i-11 at 1 μ M (lane 3). Arrows indicate expected size of wild type and mutant LMAN1 protein. **B)** Stain free loading control image for gel.

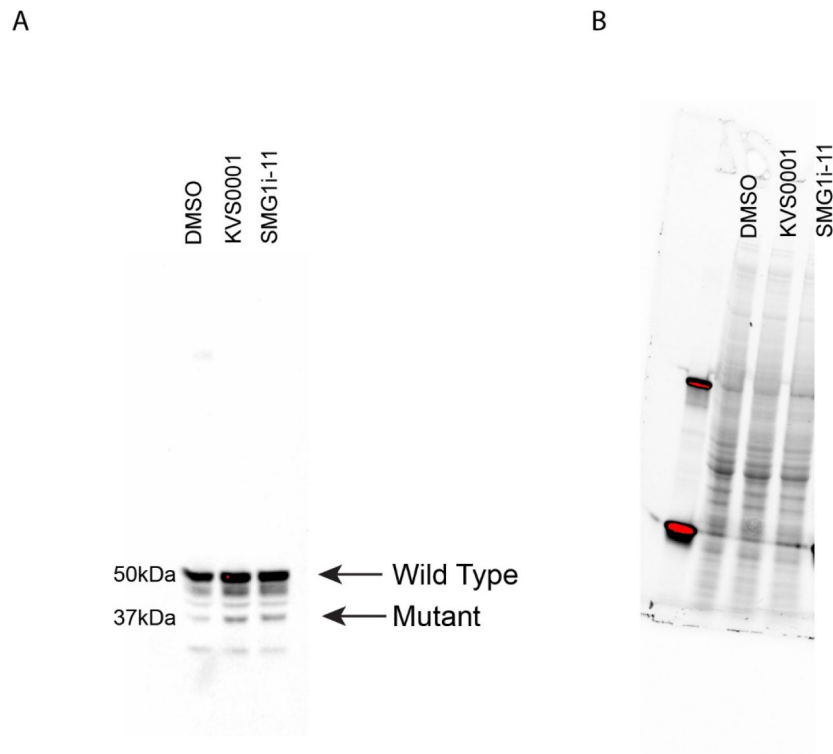


Figure S16

DNA and protein sequences for the wild type and mutant alleles of **A) EXOC1** **B) RAB14** and **C) ZDHHC16** genes. The mutant protein sequences caused by the out of frame indel are highlighted in red and the boxes indicate the peptides presented on the cell surface and identified by mass spectrometry in cells treated with 5 μ M of KVS0001 (see **Figures 4A** and **4B**).

A) EXOC1 (peptide: IIVAAHCLF)

Wild Type Allele	...TCACGGCAACA...	Translate	...TLRQHNCGTLPVSSEK...
Mutant Allele	...TCACG-CAACA...		...TLRNIIVAAHCLFHRLRKIX
	Delete "G"		

B) RAB14 (peptide: IVLTQLVLNLV)

Wild Type Allele	...AGAAAAAAAAATT...	Translate	...CLLHQFTEKKFM...
Mutant Allele	...AGAAAAAAAA-TT...		...CLLHQFTEKNLWLIIVLTQLVLNLVQEX
	Delete "A"		

C) ZDHHC16 (peptide: PLLMLPLSL)

Wild Type Allele	...TTGGGGGCAGTG...	Translate	...YNSFGGSDTAVDAA...
Mutant Allele	...TTGGGG-CAGTG...		...YNSFGAVTPLLMLPLSLSTGWX
	Delete "G"		

Figure S17

Heavy peptide loading controls and endogenous (light) peptide presentation of genes in LS180 and NCI-H358 treated with DMSO or 5µM KVS0001. Data is from quantitative HPLC-Mass Spectrometry. Note y-axis scale changes between samples. Tables on the right show relative increase in peptide presentation with KVS0001.

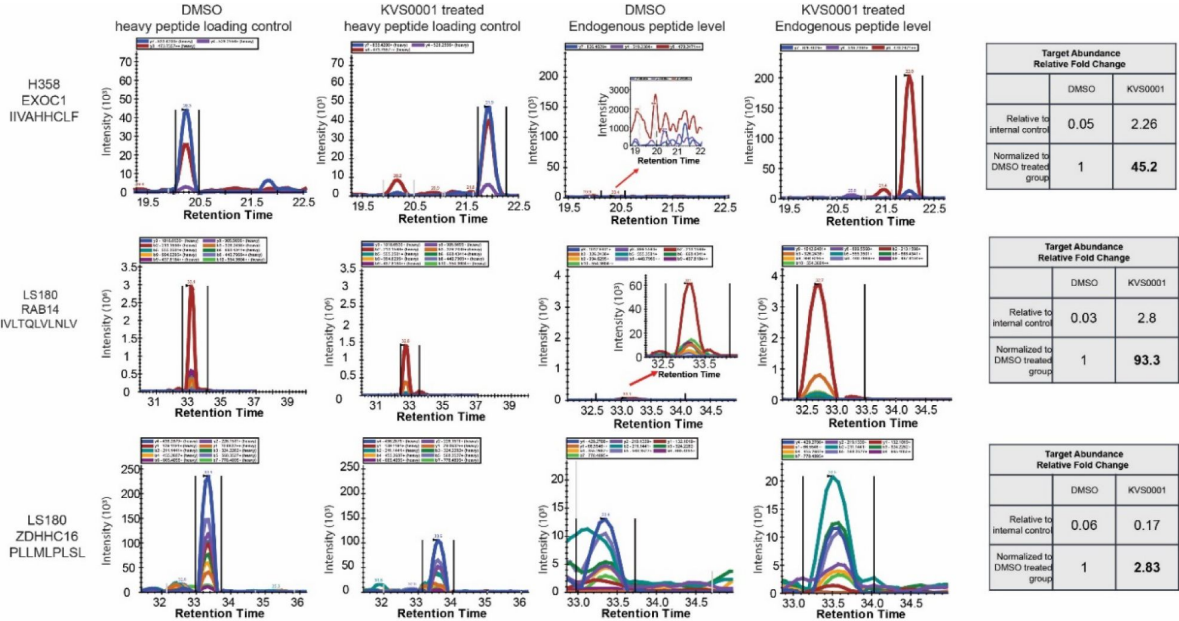


Figure S18

Waterfall plot of publicly available TP53 RNA expression (as shown by FPKM) for 675 cancer cell lines. The two cell lines used in this study are highlighted in red and were in the bottom quartile of TP53 expression.

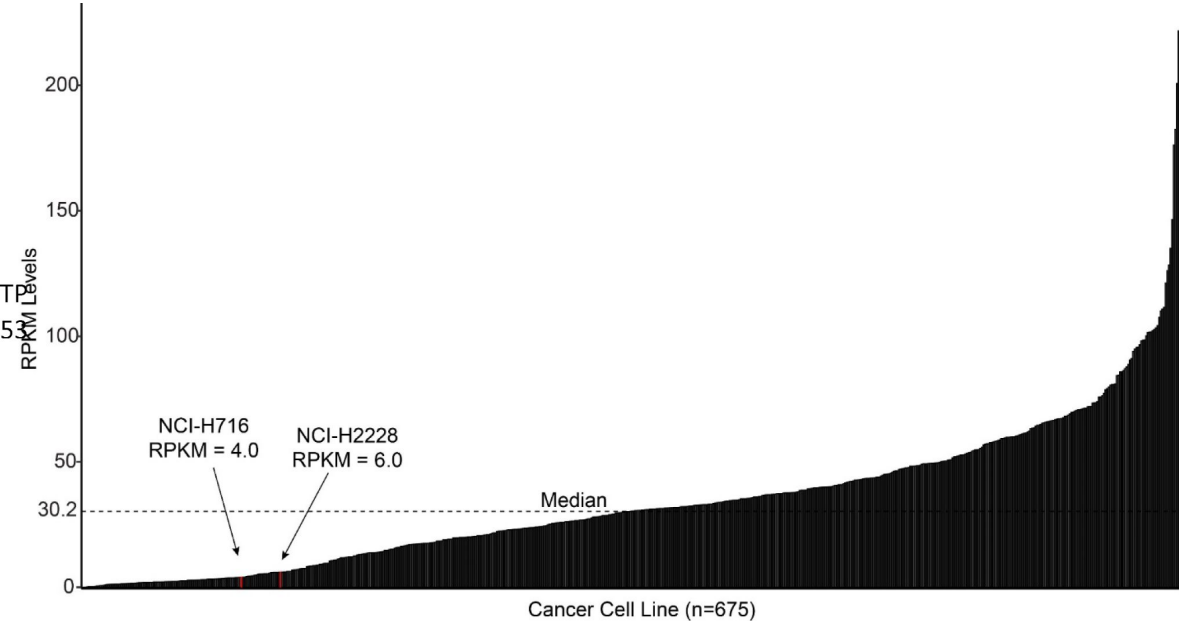


Figure S19

Western blot of TP53 on NCI-H716 and NCI-H2228 cells treated with 5 μ M or 7.5 μ M of NMD inhibitor, 1 μ M SMG1i-11, or 200mg/mL chemotherapy, showing controls related to **Figure 4D**. LYO is LY3023414 and 5-FU is 5-fluorouracil. HEK293 parent cells are shown as a control (wild type TP53 protein).

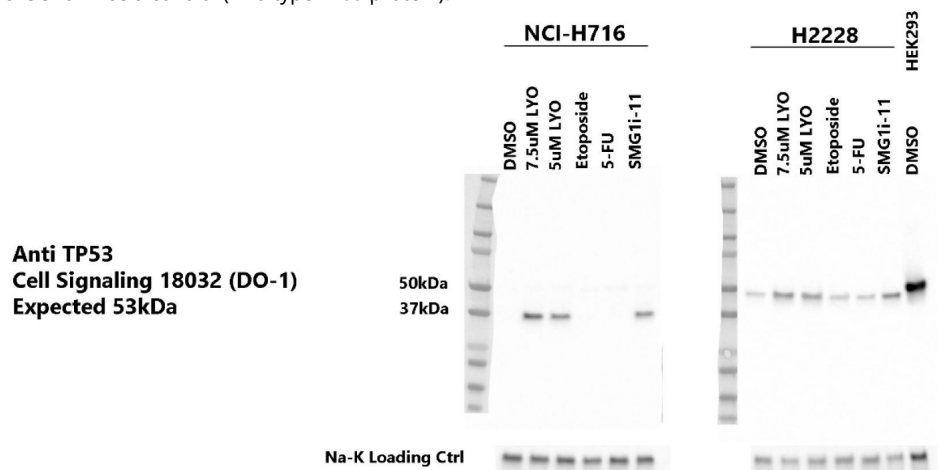
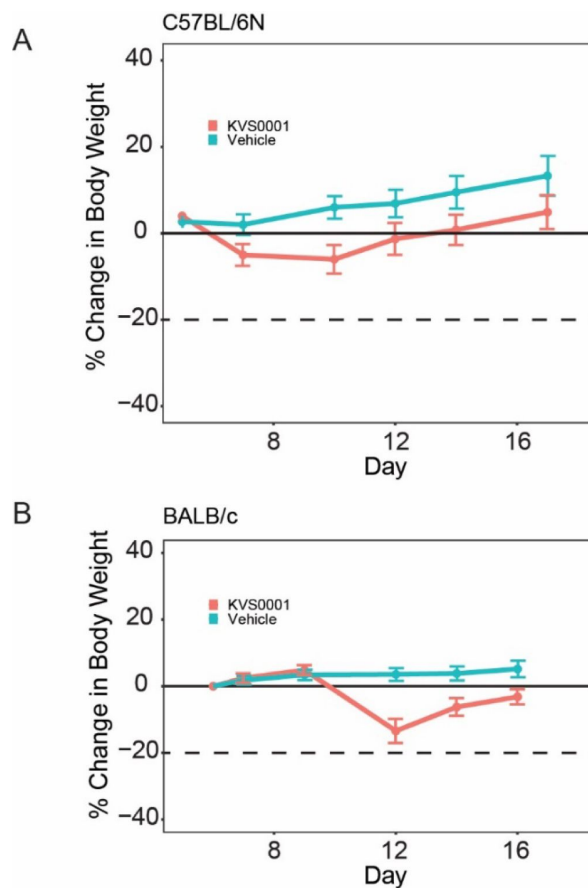
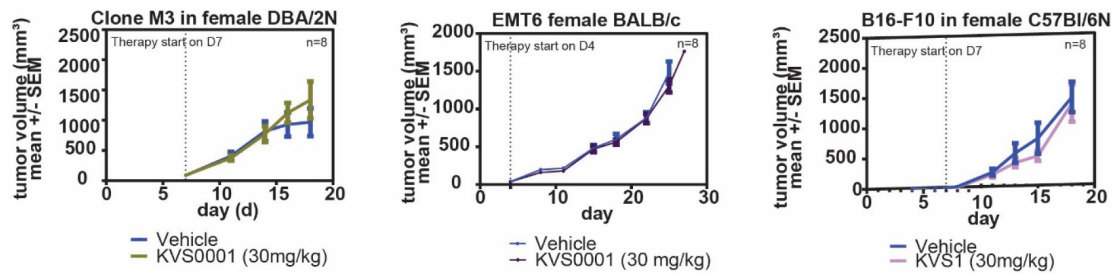


Figure S20

Mouse weights for **A)** C57BL/6N and **B)** BALB/c tumor bearing mice treated with 30mg/kg KVS0001 or vehicle control IP daily. N=8 for all arms.



A



B

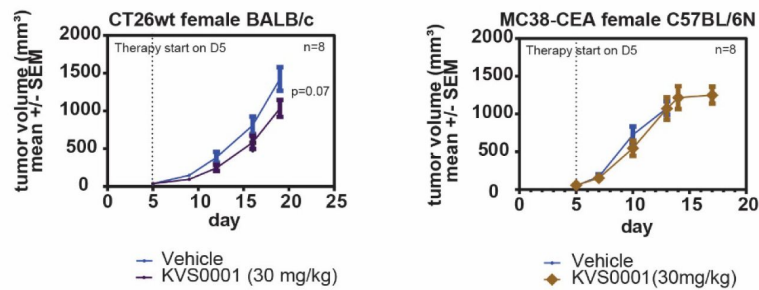


Figure S21

Mouse tumor size as measured by calipers for mammary fat pad placed syngeneic tumor models treated daily with 30mg/kg KVS0001 or vehicle. **A)** Tumors with low/moderate and **B)** high indel mutational loads are shown. No results presented in this supplemental figure are statistically significant. Error bars show 95% confidence intervals.

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Reviewer #1 (Public Review):

Summary:

This work identified new NMD inhibitors and tested them for cancer treatment, based on the hypothesis that inhibiting NMD could lead to the production of cancer neoantigens from the stabilized mutant mRNAs, thereby enhancing the immune system's ability to recognize and kill cancer cells. Key points of the study include:

- Development of an RNA-seq based method for NMD analysis using mixed isogenic cells that express WT or mutant transcripts of STAG2 and TP53 with engineered truncation mutations.
- Application of this method for a drug screen and identified several potential NMD inhibitors.
- Demonstration that one of the identified compounds, LY3023414, inhibits NMD by targeting the SMG1 protein kinase in the NMD pathway in cultured cells and mouse xenografts.
- Due to the in vivo toxicity observed for LY3023414, the authors developed 11 new SMG1 inhibitors (KVS0001-KVS0011) based on the structures of the known SMG1 inhibitor SMG1i-11 and the SMG1 protein itself.
- Among these, KVS0001 stood out for its high potency, excellent bioavailability and low toxicity in mice. Treatment with KVS0001 caused NMD inhibition and increased presentation of neoantigens on MHC-I molecules, resulting in the clearance of cancer cells in vitro by co-cultured T cells and cancer xenografts in mice by the immune system.

These findings support the strategy of targeting the NMD pathway for cancer treatment and provide new research tools and potential lead compounds for further exploration.

Strengths:

The RNA-seq based NMD analysis, using isogenic cell lines with specific NMD-inducing mutations, represents a novel approach for the high-throughput identification of potential NMD modulators or genetic regulators. The effectiveness of this method is exemplified by the identification of a new activity of AKT1/mTOR inhibitor LY3023414 in inhibiting NMD.

The properties of KVS0001 described in the manuscript as a novel SMG1 inhibitor suggest its potential as a lead compound for further testing the NMD-targeting strategies in cancer treatment. Additionally, this compound may serve as a useful research tool.

The results of the in vitro cell killing assay and in vivo xenograft experiments in both immuno-proficient and immune-deficient mice indicate that inhibiting NMD could be a viable therapeutic strategy for certain cancers.

Weaknesses:

The authors did not address the potential effects of NMD/SMG1 inhibitors on RNA splicing. Given that the transcripts of many RNA-binding proteins are natural targets of NMD, inhibiting NMD could significantly alter splicing patterns. This, in turn, might influence the outcomes of the RNA-seq-based method for NMD analysis and result interpretation.

While the RNA-seq based approach offers several advantages for analyzing NMD, the effects of NMD/SMG1 inhibitors observed through this method should be confirmed using established NMD reporters. This step is crucial to rule out the possibility that mutations in STAG2 or TP53 affect NMD in cells, as well as to address potential clonal variations between different engineered cell lines.

The results from the SMG1/UPF1 knockdown and SMG1i-11 experiments presented in Figure 3 correlate with the effects seen for LY3023414, but they do not conclusively establish SMG1 as the direct target of LY3023414 in NMD inhibition. An epistatic analysis with LY3023414 and SMG1-knockdown is needed.

Comment on the revised version:

Although KVS0001 exhibits promising properties as an SMG1 inhibitor for cancer treatment, it remains unclear if it is superior to existing SMG1 inhibitors, as no direct comparisons have been made.

<https://doi.org/10.7554/eLife.95952.2.sa2>

Reviewer #2 (Public Review):

Summary:

Several publications during the past years provided evidence that NMD protects tumor cells from being recognized by the immune system by suppressing the display of neoantigens, and hence NMD inhibition is emerging as a promising anti-cancer approach. However, the lack of an efficacious and specific small molecule NMD inhibitor with suitable pharmacological properties is currently a major bottleneck in the development of therapies that rely on NMD inhibition. In this manuscript, the authors describe their screen for identifying NMD inhibitors, which is based on isogenic cell lines that either express wild-type or NMD-sensitive transcript isoforms of p53 and STAG2. Using this setup, they screened a library of

2658 FDA-approved or late-phase clinical trial drugs and had 8 hits. Among them they further characterized LY3023414, showing that it inhibits NMD in cultured cells and in a mouse xenograft model, where it, however, was very toxic. Because LY3023414 was originally developed as a PI3K inhibitor, the authors claim that it inhibits NMD by inhibiting SMG1. While this is most likely true, the authors do not provide experimental evidence for this claim. Instead, they use this statement to switch their attention to another previously developed SMG1 inhibitor (SMG1i-11), of which they design and test several derivatives. Of these derivatives, KVS0001 showed the best pharmacological behavior. It upregulated NMD-sensitive transcripts in cultured cells and the xenograft mouse model, and two predicted neoantigens could indeed be detected by mass spectrometry when the respective cells were treated with KVS0001. A bispecific antibody targeting T cells to a specific antigen-HLA complex led to increased IFN-gamma release and killing of cancer cells expressing this antigen-HLA complex when they were treated with KVS0001. Finally, the authors show that renal (RENCA) or lung cancer cells (LLC) were significantly inhibited in tumor growth in immunocompetent mice treated with KVS0001. Overall, this establishes KVS0001 as a novel and promising ant-cancer drug that by inhibiting SMG1 (and therewith NMD) increases the neoantigen production in the cancer cells and reveals them to the body's immune system as "foreign".

Strengths:

The novelty and significance of this work consist in the development of a novel and - judging from the presented data - very promising NMD inhibiting drug that is suitable for applications in animals. This is an important advance for the field, as previous NMD inhibitors were not specific, lacked efficacy, or were very toxic and hence not suitable for animal application. It will be still a long way with many challenges ahead towards an efficacious NMD inhibitor that is safe for use in humans, but KVS0001 appears to be a molecule that bears promise for follow-up studies. In addition, while the idea of inhibiting NMD to trigger neoantigen production in cancer cells and so reveal them to the immune system has been around for quite some time, this work provides ample and compelling support for the feasibility of this approach, at least for tumors with a high mutational burden.

Main weaknesses:

There is a disconnect between the screen and the KVS0001 compound, that they describe and test in the second part of the manuscript since KVS0001 is a derivative of the SMG1 inhibitors developed by Gopalsamy et al. in 2012 and not of the lead compound identified in the screen (LY3023414). Because of high toxicity in the mouse xenograft experiments, the authors did not follow up LY3023414 but instead switched to the published SMG1i-11 drug of Gopalsamy and colleagues, a molecule that is widely used among NMD researchers for NMD inhibition in cultured cells. Therefore, in my view, the description of the screen is obsolete, and the paper could just start with the optimization of the pharmacological properties of SMG1i-11 and the characterization of KVS0001. Even though the screen is based on an elegant setup and was executed successfully, it was ultimately a failure as it didn't reveal a useful lead compound that could be further optimized.

Additional points:

- Compared to SMG1i-11, KVS0001 seems less potent in inhibiting SMG1 (higher IC₅₀). It would therefore be important to also compare the specificity of both drugs for SMG1 over other kinases at the actually applied concentrations (1 μ M for SMG1i-11, 5 μ M for KVS0001). The Kinativ Assay (Fig. S13) was performed with 100 nM KVS0001, which is 50-fold less than the concentration used for functional assays and hence not really meaningful. In addition, more information on the pharmacokinetic properties and toxicology of KVS0001 would allow a better judgment of the potential of this molecule as a future therapeutic agent.
- On many figures, the concentrations of the used drugs are missing. Please ensure that for

every experiment that includes drugs, the drug concentration is indicated.

- Do the authors have an explanation for why LY3023414 has a much stronger effect on the p53 than on the STAG2 nonsense allele (Fig. 1B, S8), whereas emetine upregulates the STAG2 nonsense alleles more than the p53 nonsense allele (Fig. S5). I find this curious, but the authors do not comment on it.

- While it is a strength of the study that the NMD inhibitors were validated on many different truncation mutations in different cell lines, it would help readers if a table or graphic illustration was included that gives an overview of all mutant alleles tested in this study (which gene, type of mutation, in which cell type). In the current version, this information is scattered throughout the manuscript.

- Lines 194 and 302: That SMG1i-11 was highly insoluble in the hands of the authors is surprising. It is unclear why they used variant 11j, since variant 11e of this inhibitor is widely used among NMD researchers and readily dissolves in DMSO.

- Line 296: The authors claim that they were able to show that LY3023414 inhibited the SMG1 kinase, which is not true. To show this, they would have for example to show that LY3023414 prevents SMG1-mediated UPF1 phosphorylation, as they did for KVS0001 and SMG1i-11 in Fig. 3F. Unless the authors provide this data, the statement should be deleted or modified.

Comments on the revised version:

- The authors have satisfactorily addressed all my "Additional points" listed above.

- With the new publishing model of Life, the authors ultimately decide on whether or not to follow reviewers suggestions, and in this case, the authors decided (against my suggestion) to leave the screening part in the manuscript, although it did not result in a useful lead compound. They argue it helped them define in an unbiased way SMG1 as the ideal target for NMD disruption. I would counterargue that this has been known in the field for quite a while.

- One last suggestion I have to the authors would be to modify the statement in the abstract "This led to the design of a novel SMG1 inhibitor", because what they call "novel" is, in reality, a chemical improvement of the pharmacological properties of a previously reported SMG1 inhibitor (Gopalsamy et al., 2012).

<https://doi.org/10.7554/eLife.95952.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

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This is a very important comment that highlights an important aspect of NMD and potential exciting downstream studies. We did not systematically assess RNA splicing in our work as we are not sure if inhibition of NMD would induce cancer specific splicing that would allow for tumor targeting. It is well established that NMD can impact splicing, including modulating cryptic exon expression, but finding and assessing antigenicity of targetable tumor specific antigens constitutes a study in and of its own. Our own data in figure 4C-F supports this, as a point mutation near a splice site in TP53 strongly induced NMD which was subsequently stopped by KVS0001 treatment. Doing a systematic review of this effect we feel is outside the scope of this manuscript. We've incorporated a comment into our discussion highlighting this deficiency, but certainly find the idea of mining RNA-splicing changes an exciting next endeavor.

While the RNA-seq-based approach offers several advantages for analyzing NMD, the effects of NMD/SMG1 inhibitors observed through this method should be confirmed using established NMD reporters. This step is crucial to rule out the possibility that

mutations in STAG2 or TP53 affect NMD in cells, as well as to address potential clonal variations between different engineered cell lines.

This is possible, but we want to highlight that all hits from the screen were confirmed in a separate cell line with different clones. While this will not rule out effects to NMD due to STAG2 and TP53 knockdown, the final lead compound was also tested on different endogenous transcripts in both indel and normal transcripts controlled by NMD (i.e., ATF4) in multiple species (human and mouse). Importantly, many of these assays employed the non-mutated transcripts from heterozygous mutant cells to ensure that cis-acting NMD was being measured and to control for any trans-acting splicing or other unanticipated biochemical effects.

The results from the SMG1/UPF1 knockdown and SMG1i-11 experiments presented in Figure 3 correlate with the effects seen for LY3023414, but they do not conclusively establish SMG1 as the direct target of LY3023414 in NMD inhibition. An epistatic analysis with LY3023414 and SMG1-knockdown is needed.

This is a great comment, and is supported by the recent push to confirm drug targets by chemical probes or knockout followed by loss of further effect due to the application of the drug in question. We attempted to knockout SMG1 in multiple cells lines used in this study, including RPE1, MCF10A, NCI-H358 and LS180, and were unable to obtain clones that have biallelic out of frame indels. We were able to obtain multiple clones with in frame indels. Based on our results and those in the publicly available database DepMap we suspect this gene is likely essential, making a simple knockout unfeasible. While this uncertainty is important to keep in mind, we feel it does not detract from the reporting of a novel NMD screen that is mechanistically agnostic and of a novel *in vivo* active NMD inhibitor.

Reviewer #2 (Public Review):

Summary:

Several publications during the past years provided evidence that NMD protects tumor cells from being recognized by the immune system by suppressing the display of neoantigens, and hence NMD inhibition is emerging as a promising anti-cancer approach. However, the lack of an efficacious and specific small-molecule NMD inhibitor with suitable pharmacological properties is currently a major bottleneck in the development of therapies that rely on NMD inhibition. In this manuscript, the authors describe their screen for identifying NMD inhibitors, which is based on isogenic cell lines that either express wild-type or NMD-sensitive transcript isoforms of p53 and STAG2. Using this setup, they screened a library of 2658 FDA-approved or late-phase clinical trial drugs and had 8 hits. Among them they further characterized LY3023414, showing that it inhibits NMD in cultured cells and in a mouse xenograft model, where it, however, was very toxic. Because LY3023414 was originally developed as a PI3K inhibitor, the authors claim that it inhibits NMD by inhibiting SMG1. While this is most likely true, the authors do not provide experimental evidence for this claim. Instead, they use this statement to switch their attention to another previously developed SMG1 inhibitor (SMG1i-11), of which they design and test several derivatives. Of these derivatives, KVS0001 showed the best pharmacological behavior. It upregulated NMD-sensitive transcripts in cultured cells and the xenograft mouse model and two predicted neoantigens could indeed be detected by mass spectrometry when the respective cells were treated with KVS0001. A bispecific antibody targeting T cells to a specific antigen-HLA complex led to increased IFN-gamma release and killing of cancer cells expressing this antigen-HLA complex when they were treated with KVS0001. Finally, the authors show that renal (RENCA) or lung cancer cells (LLC) were significantly inhibited in tumor growth in immunocompetent mice treated with KVS0001. Overall, this establishes KVS0001 as a novel and promising ant-cancer

drug that by inhibiting SMG1 (and therewith NMD) increases the neoantigen production in the cancer cells and reveals them to the body's immune system as "foreign".

Strengths:

The novelty and significance of this work consists in the development of a novel and - judging from the presented data - very promising NMD inhibiting drug that is suitable for applications in animals. This is an important advance for the field, as previous NMD inhibitors were not specific, lacked efficacy, or were very toxic and hence not suitable for animal application. It will be still a long way with many challenges ahead towards an efficacious NMD inhibitor that is safe for use in humans, but KVS0001 appears to be a molecule that bears promise for follow-up studies. In addition, while the idea of inhibiting NMD to trigger neoantigen production in cancer cells and so reveal them to the immune system has been around for quite some time, this work provides ample and compelling support for the feasibility of this approach, at least for tumors with a high mutational burden.

Main weaknesses:

There is a disconnect between the screen and the KVS0001 compound, that they describe and test in the second part of the manuscript since KVS0001 is a derivative of the SMG1 inhibitors developed by Gopalsamy et al. in 2012 and not of the lead compound identified in the screen (LY3023414). Because of high toxicity in the mouse xenograft experiments, the authors did not follow up LY3023414 but instead switched to the published SMG1i-11 drug of Gopalsamy and colleagues, a molecule that is widely used among NMD researchers for NMD inhibition in cultured cells. Therefore, in my view, the description of the screen is obsolete, and the paper could just start with the optimization of the pharmacological properties of SMG1i-11 and the characterization of KVS0001. Even though the screen is based on an elegant setup and was executed successfully, it was ultimately a failure as it didn't reveal a useful lead compound that could be further optimized.

This is a helpful observation from an outside perspective. From our point of view, we were only alerted to the targeting SMG1 due to the previously reported off-target effects of LY3023414 on SMG and lack of plausible explanation for PIK3CA inhibition to efficiently inhibit NMD. We do feel that the screen is worth including for two reasons. First, it offers an unbiased approach for querying the entire NMD pathway for vulnerabilities useful to target. The library chosen was quite small, so the screen itself could be useful to others with larger libraries to test. Second, it did help identify SMG1 as the ideal target for NMD disruption. While targeting SMG1 is not novel, we felt it highlighted why we chose to develop KVS0001. To address this reviewer's comment, we've included a couple sentences in the results and discussion strengthening the point that the screen provided an unbiased approach to finding the best target in the pathway to disrupt NMD and elaborating on the transition from LY3023414 and the screen to development of KVS0001.

Additional points:

- Compared to SMG1i-11, KVS0001 seems less potent in inhibiting SMG1 (higher IC₅₀). It would therefore be important to also compare the specificity of both drugs for SMG1 over other kinases at the applied concentrations (1 μ M for SMG1i-11, 5 μ M for KVS0001). The Kinativ Assay (Fig. S13) was performed with 100 nM KVS0001, which is 50-fold less than the concentration used for functional assays and hence not really meaningful. In addition, more information on the pharmacokinetic properties and toxicology of KVS0001 would allow a better judgment of the potential of this molecule as a future therapeutic agent.

We agree that the Kinativ assay may have poorly represented the activity of KVS0001 at the bioactive concentration. We have now added 1uM Kinativ data, the highest concentration we were able to run to figure S13.

- On many figures, the concentrations of the used drugs are missing. Please ensure that for every experiment that includes drugs, the drug concentration is indicated.

We apologize for this oversight and have added all drug concentrations on the appropriate plots.

- Do the authors have an explanation for why LY3023414 has a much stronger effect on the p53 than on the STAG2 nonsense allele (Figure 1B, S8), whereas emetine upregulates the STAG2 nonsense alleles more than the p53 nonsense allele (Figure S5). I find this curious, but the authors do not comment on it.

This is an interesting observation. The short answer is we're not sure. The speculative answer is that it is related to the distinctly different mechanisms of actions of the two inhibitors (see comments from reviewing editor below).

- While it is a strength of the study that the NMD inhibitors were validated on many different truncation mutations in different cell lines, it would help readers if a table or graphic illustration was included that gives an overview of all mutant alleles tested in this study (which gene, type of mutation, in which cell type). In the current version, this information is scattered throughout the manuscript.

This is an excellent suggestion. We've included a new table S1 which incorporates the details of each cell line and the genes used in each for this study.

- Lines 194 and 302: That SMG1i-11 was highly insoluble in the hands of the authors is surprising. It is unclear why they used variant 11j, since variant 11e of this inhibitor is widely used among NMD researchers and readily dissolves in DMSO.

As this referee notes SMG1i-11 is soluble in DMSO in our hands as well, which enabled us to use it for our *in vitro* work. Unfortunately, the concentrations of DMSO required to dissolve the compound to suitable concentrations for *in vivo* work were too high to safely use in mice with our animal protocols. We also attempted to use ethanol, which also did dissolve SMG1i-11, but led to a significant amount of toxicity in both the drug and vehicle control arms.

- Line 296: The authors claim that they were able to show that LY3023414 inhibited the SMG1 kinase, which is not true. To show this, they would have for example to show that LY3023414 prevents SMG1-mediated UPF1 phosphorylation, as they did for KVS0001 and SMG1i-11 in Fig. 3F. Unless the authors provide this data, the statement should be deleted or modified.

We've modified this statement as requested by the referee, now saying we suspected SMG1 was the target based on previously published work.

Recommendations for the authors:

Reviewing Editor (Recommendations For The Authors):

*Your paper has been assessed by two reviewers with expertise in the NMD field. They both find the identification and characterization of a new potent and selective inhibitor of the SMG1 NMD kinase with *in vivo* activity to represent a significant advance in the*

field, and one that could ultimately be of value as the basis for a novel cancer therapy. However, as you will see both reviewers have concerns about whether the SMG1 inhibitor screen you developed belongs in the paper because it was not used to identify the KVS0001 inhibitor, which instead was generated based on a previously published set of SMG1 inhibitors, and because the NMD inhibitor that did emerge from your screen, LY3023414, was not shown to be a direct inhibitor of SMG1 kinase activity. While it is an elegant screen, during the revision of the paper you could consider streamlining the manuscript by emphasizing how the screening assay was used to validate KVS0001, and bolstering the characterization of the new KVS0001 NMD inhibitor by conducting the proposed additional experiments.

Each of the reviewers raises additional points that should be addressed in a revised version.

The reviewing editor has two additional points:

(1) While emetine inhibits NMD, it is not really a direct NMD inhibitor, as implied, but rather a potent protein synthesis elongation inhibitor that acts by binding to the E-site of the 40S ribosomal subunit, and is therefore, like anisomycin, another protein synthesis inhibitor, working indirectly to inhibit NMD. This should be acknowledged in the section where emetine is first used as an "NMD inhibitor".

This has been included in the indicated section at the referee's request.

(2) To establish that the observed phenotypic effects of KVS0001 are due to on-target inhibition of SMG1, the authors could generate and express an SMG1 point mutant that is resistant to KVS0001 inhibition, which could be based on the SMG1 catalytic domain structure that the authors used originally to design KVS001. Inhibitor-resistant kinase mutants are the gold standard for demonstrating that the biological consequences of a novel protein kinase inhibitor are due to on-target effects. Admittedly, because SMG1 is such a huge protein, this may be technically challenging and is likely beyond the scope of the present paper.

-We agree with the reviewing editor on all accounts: this would be an ideal experiment to run, but also that it is beyond the scope of the present paper. As indicated in our discussion above with reviewer 1, SMG1 knockout was not possible in our hands, and we suspect it may be due to the gene being essential. Creating an inhibitor resistant mutant could overcome this issue and create an ideal model to test the target for KVS0001. Unfortunately finding such a mutant would likely require significant amounts of trial and error to create a resistant mutant that did not lose SMG1 function. And SMG1 is huge, creating technical issues for experimenting. Due to the anticipated amount of work for such a study we believe this would be better accomplished in future studies.

Reviewer #1 (Recommendations For The Authors):

(1) The authors did not mention a new SMG1 inhibitor and its effects described in Cheruiyot et al, Cancer Res 2019 (PMID: 34215620).

A comment regarding this discovery and its implications for our work was added to the discussion.

(2) There is an inconsistency between the manuscript text and methods sections regarding the time of drug treatment (16 hours vs 14 hours) in the HTS screen.

This has been double checked in our notebook and fixed to reflect 16hrs as the correct incubation time. Thank you for identifying that clerical oversight.

Reviewer #2 (Recommendations For The Authors):

(1) Line 61: The references to NMD reviews are very old (refs 20 and 21). I suggest citing more recent, up-to-date reviews instead.

Two additional references, one from 2016 and another from 2023, have been added to increase support for this statement in the introduction.

(2) Figure S1: Shouldn't the caption of the right panel (TP32 data) say "clone 221" rather than "clone 22"?

This has been fixed.

(3) Figure S18: Please indicate on the y-axis that you are displaying RPKM for p53.

This has been fixed.

(4) Figures 4D and S19: Please indicate concentrations used for all drugs.

This has been fixed.

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